

Why Public-Safety Wages May Grow Differently

A Visual Atlas of Municipal Compensation Evidence

Compensation mechanisms, claim boundaries, and project-wide limitations

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Gabriel Wages · August 2026

What this atlas can—and cannot—show

38

compensation mechanism categories

14

final adjudicated claims

7

retained counterexamples

201

unresolved high-impact conflicts

0

compatible external wage matches

0

compatible external growth pairs

I found extensive evidence about how municipal compensation is negotiated, adopted, implemented, and constrained. The strongest result is a bounded mechanism account, not a national wage-gap estimate. This atlas shows every documented mechanism, all final claims, the evidence that cuts against them, and the limits that prevented stronger conclusions.

How to read the maps and evidence tiers

-  **Strict evidence** Exact, claim-safe evidence with compatible units and source support.
-  **Bounded evidence** Usable comparisons with an explicit role, period, or source caveat.
-  **Directional evidence** Evidence about mechanism or direction without a defensible magnitude.
-  **Context** Relevant background that does not directly support a claim.
-  **Rejected** Evidence that cannot support the proposed interpretation.

A hex map groups nearby municipalities into equal-sized areas so regional patterns are easier to see. The shading represents documented compensation events, not population or national prevalence. All mechanism maps use deduplicated municipality × compensation cycle × mechanism × side implementation events.

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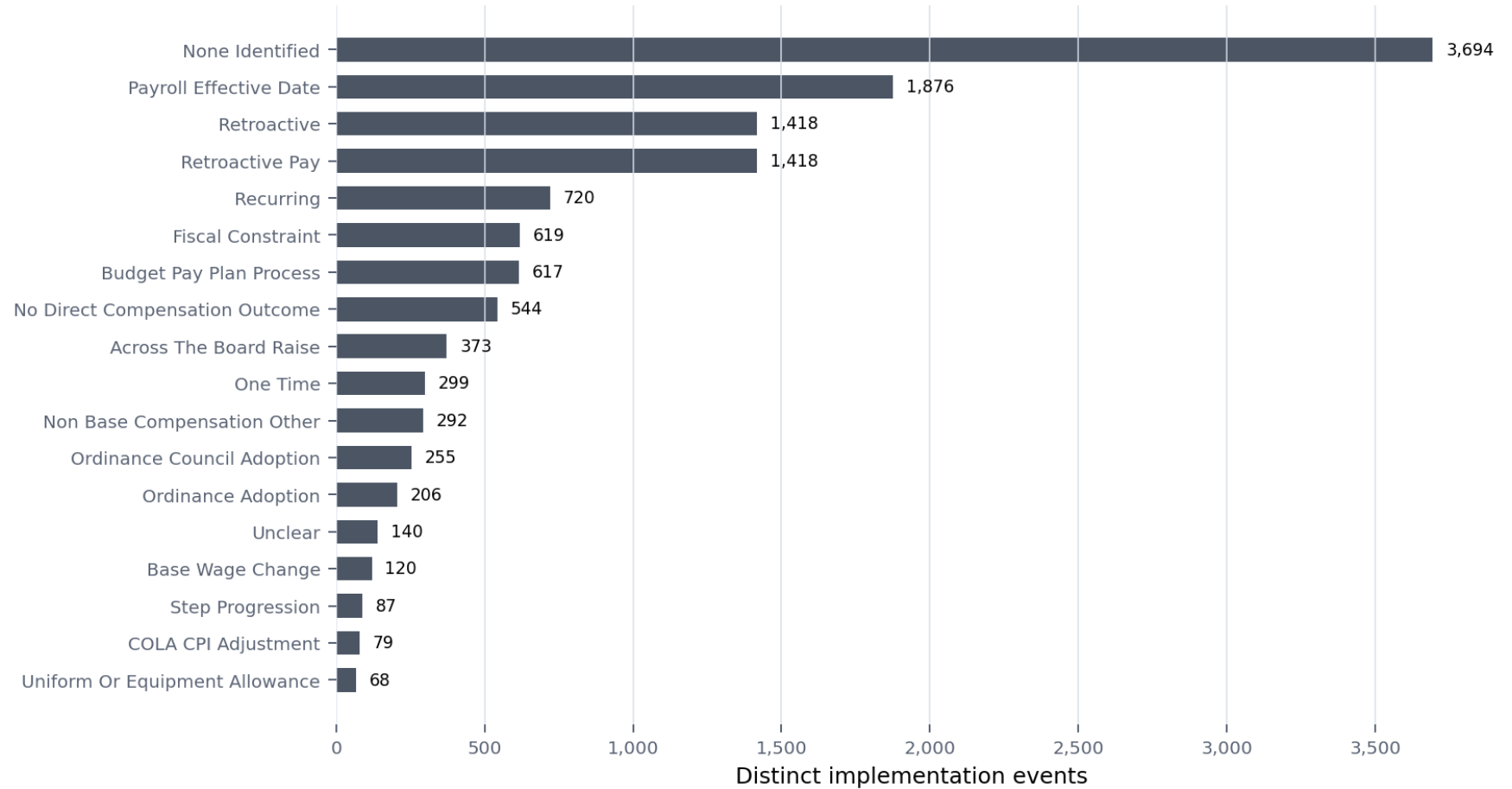
Section I

Compensation mechanisms

A visual section built from adjudicated, source-linked evidence.

Compensation mechanisms in the retained evidence

The 18 most frequently documented categories; all 38 appear in the atlas

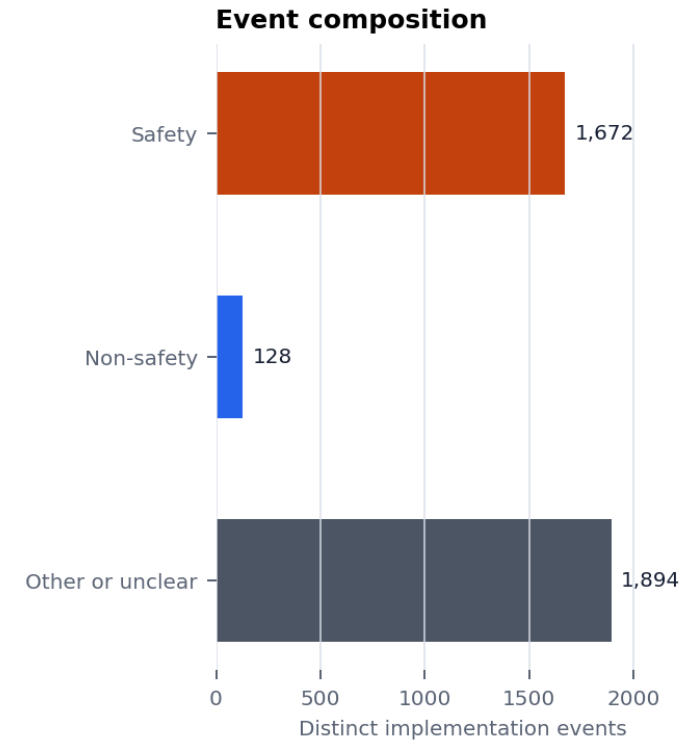
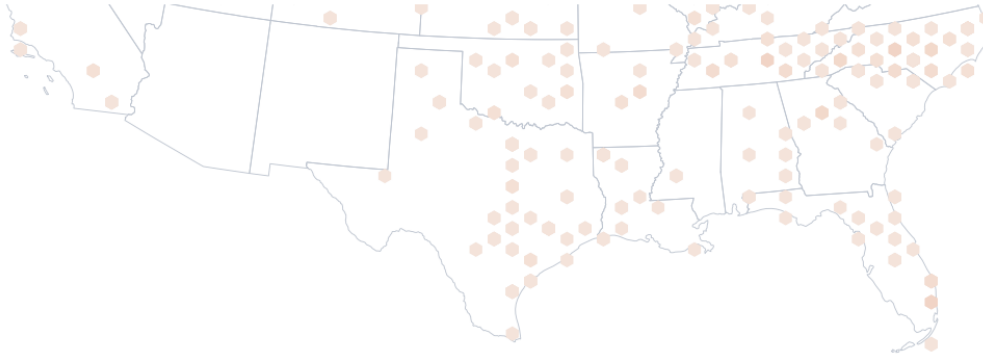


Counts use deduplicated municipality × compensation cycle × mechanism × side implementation events. Frequency in this corpus is not national prevalence.

This overview preserves the final analytical unit and evidence boundary. Counts show documented evidence in this project, not national prevalence or causal effects.

None Identified

3,694 deduplicated implementation events · 1,156 municipalities · 42 states



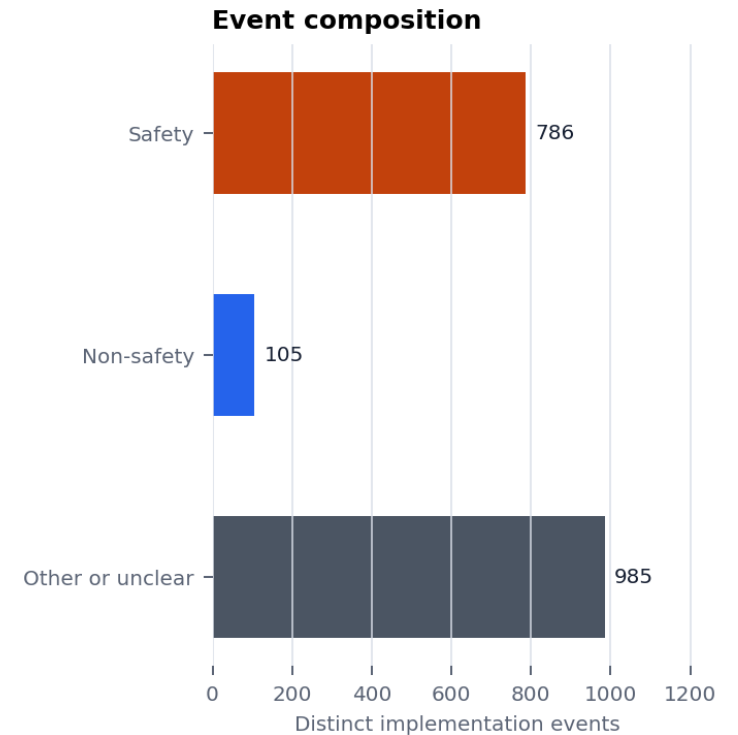
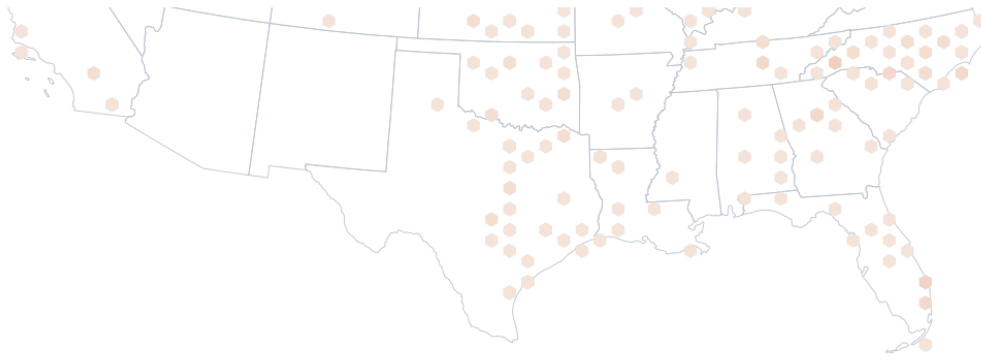
No specific compensation mechanism was identified in the retained event record.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 3,694 distinct documented none identified implementation events across 1,156 municipalities in 42 states. No specific compensation mechanism was identified in the retained event record. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Payroll Effective Date

1,876 deduplicated implementation events · 1,119 municipalities · 42 states



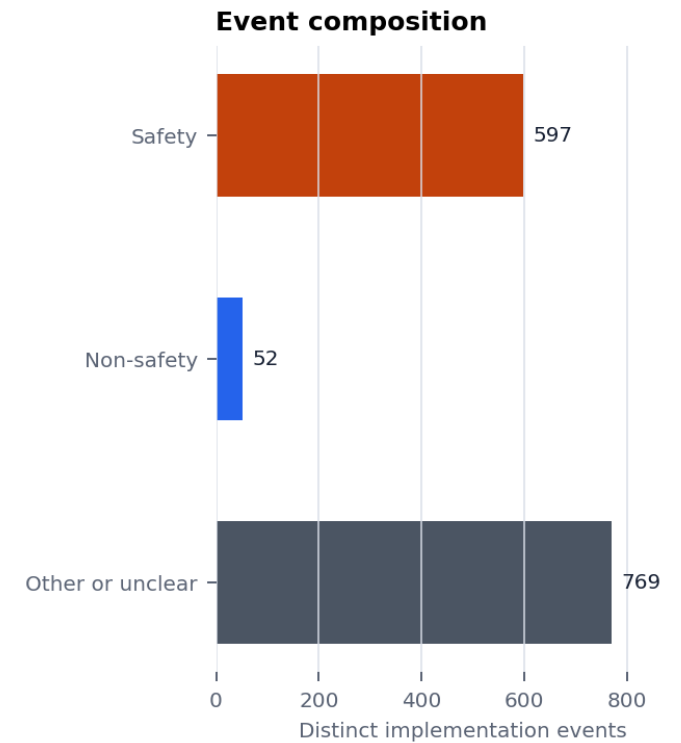
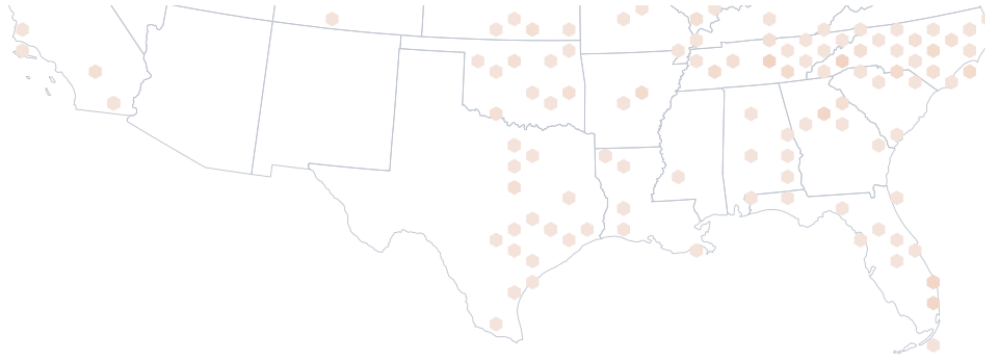
The date a compensation change became operative in payroll administration.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 1,876 distinct documented payroll effective date implementation events across 1,119 municipalities in 42 states. The date a compensation change became operative in payroll administration. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Retroactive

1,418 deduplicated implementation events · 922 municipalities · 42 states



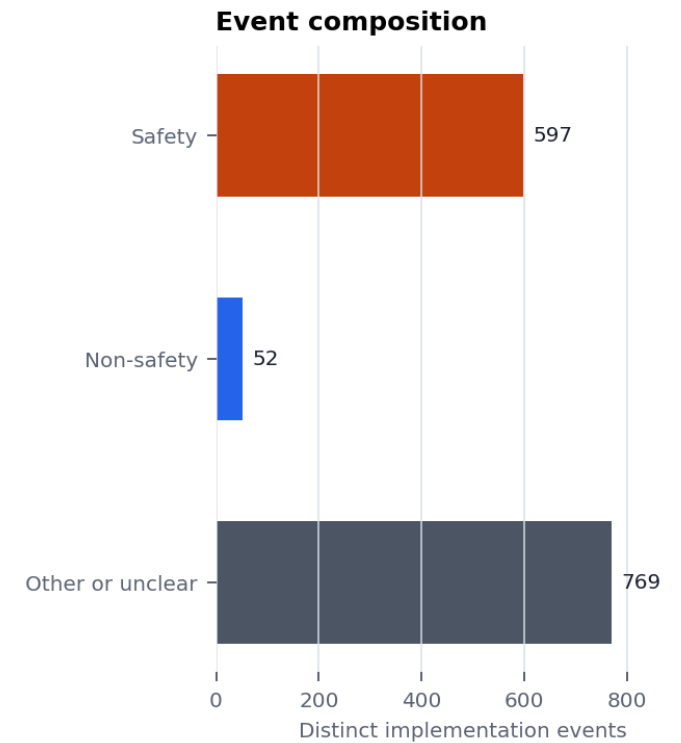
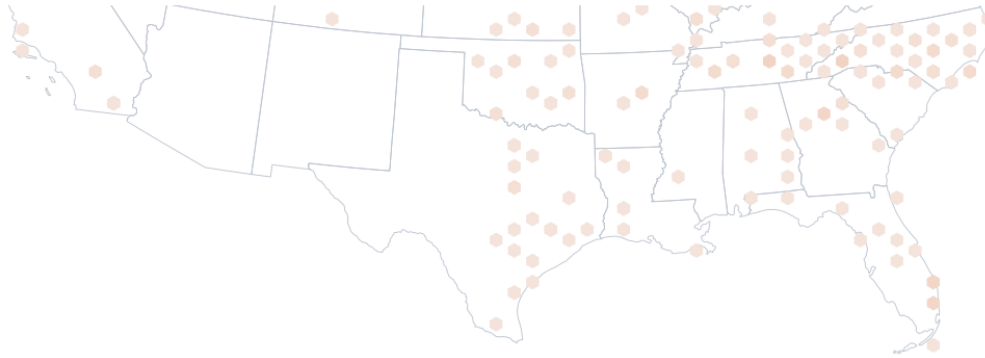
A compensation change applied to an earlier effective period.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 1,418 distinct documented retroactive implementation events across 922 municipalities in 42 states. A compensation change applied to an earlier effective period. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Retroactive Pay

1,418 deduplicated implementation events · 922 municipalities · 42 states



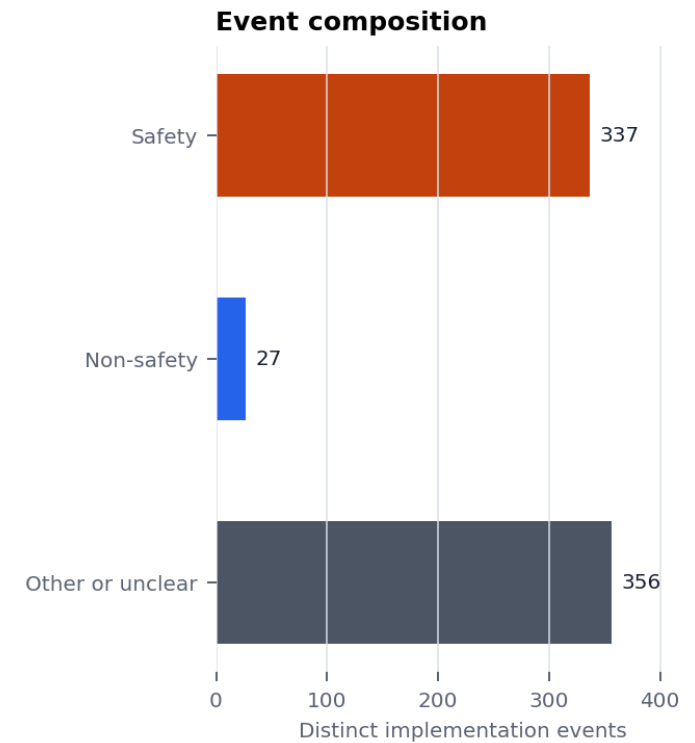
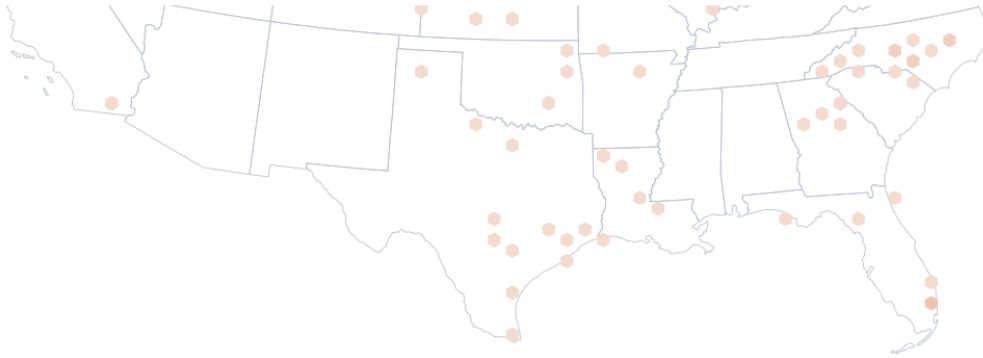
Back pay issued after a compensation term took effect or was settled.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 1,418 distinct documented retroactive pay implementation events across 922 municipalities in 42 states. Back pay issued after a compensation term took effect or was settled. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Recurring

720 deduplicated implementation events · 515 municipalities · 35 states



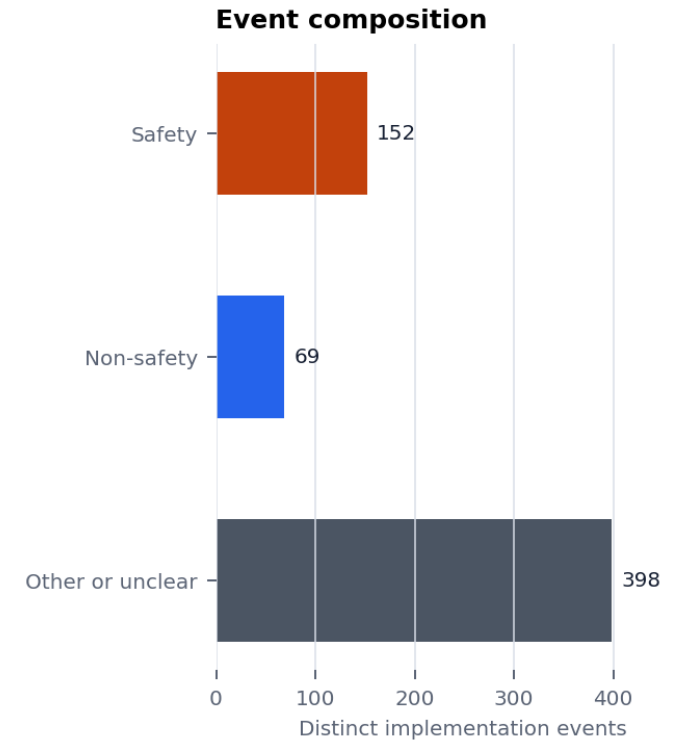
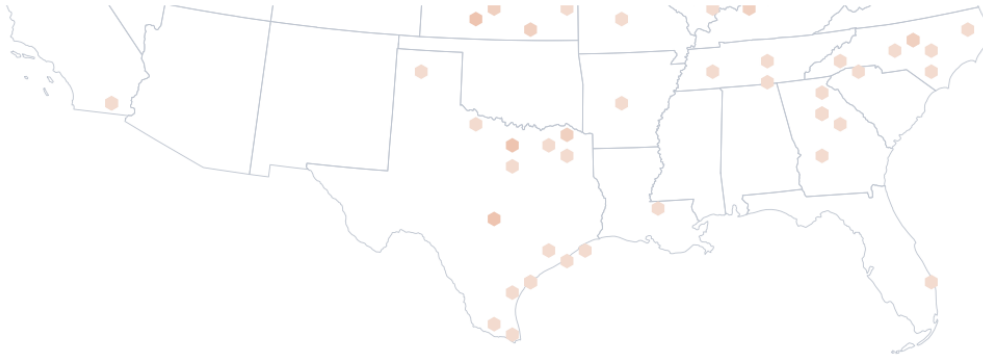
A compensation component that continues rather than ending after one payment.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 720 distinct documented recurring implementation events across 515 municipalities in 35 states. A compensation component that continues rather than ending after one payment. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Fiscal Constraint

619 deduplicated implementation events · 395 municipalities · 33 states



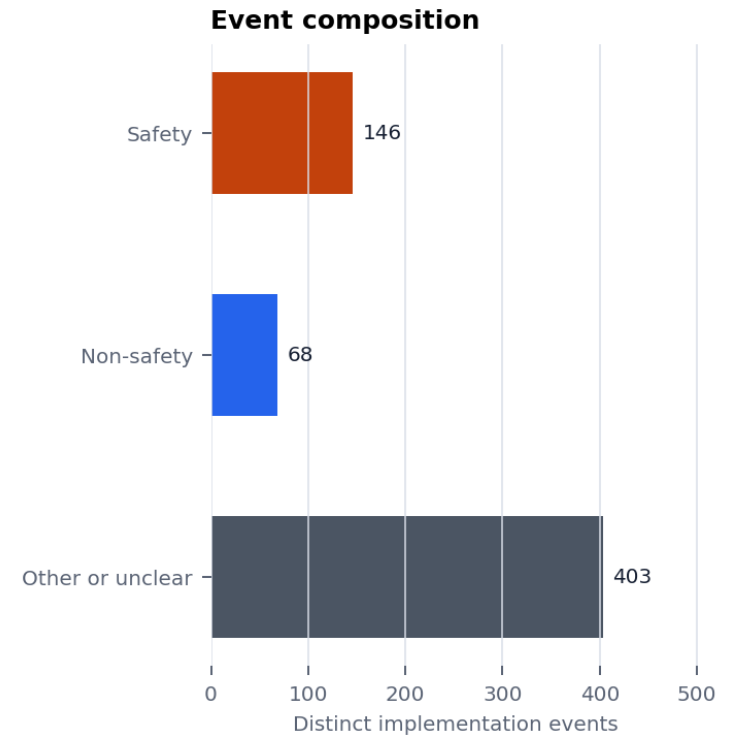
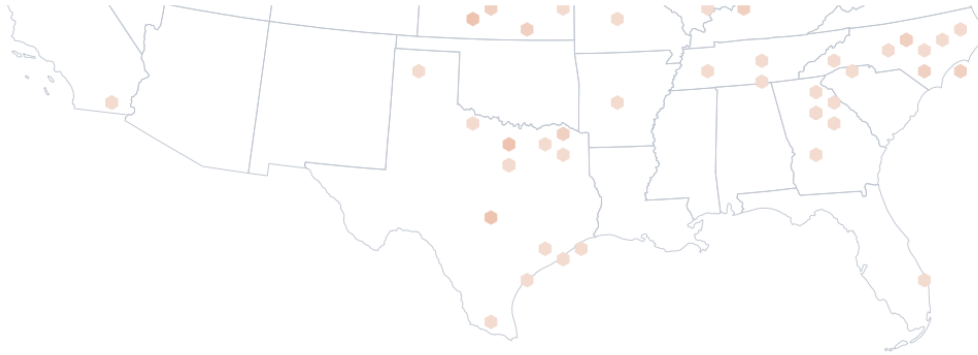
Budget pressure or limited fiscal capacity that constrained compensation choices.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 619 distinct documented fiscal constraint implementation events across 395 municipalities in 33 states. Budget pressure or limited fiscal capacity that constrained compensation choices. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Budget Pay Plan Process

617 deduplicated implementation events · 391 municipalities · 31 states



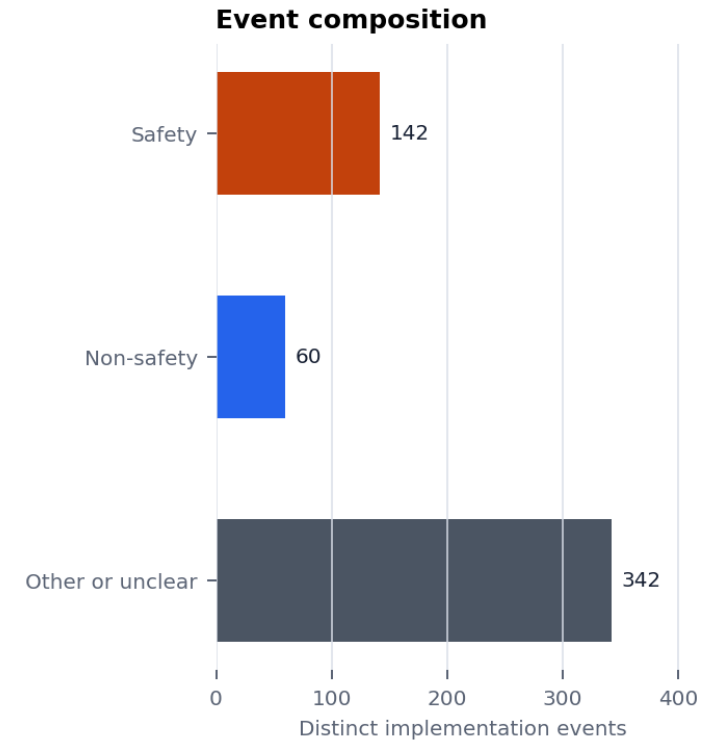
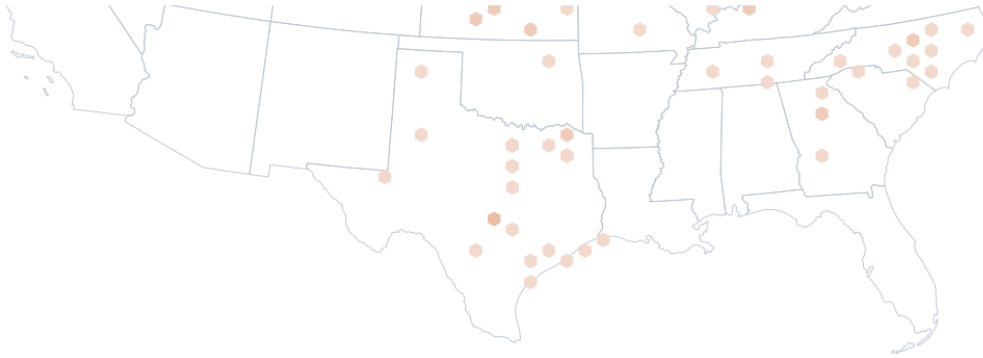
A budget or pay-plan process that formally set or revised compensation.

Analytical unit: municipality x compensation cycle x mechanism x side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 617 distinct documented budget pay plan process implementation events across 391 municipalities in 31 states. A budget or pay-plan process that formally set or revised compensation. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

No Direct Compensation Outcome

544 deduplicated implementation events · 375 municipalities · 28 states



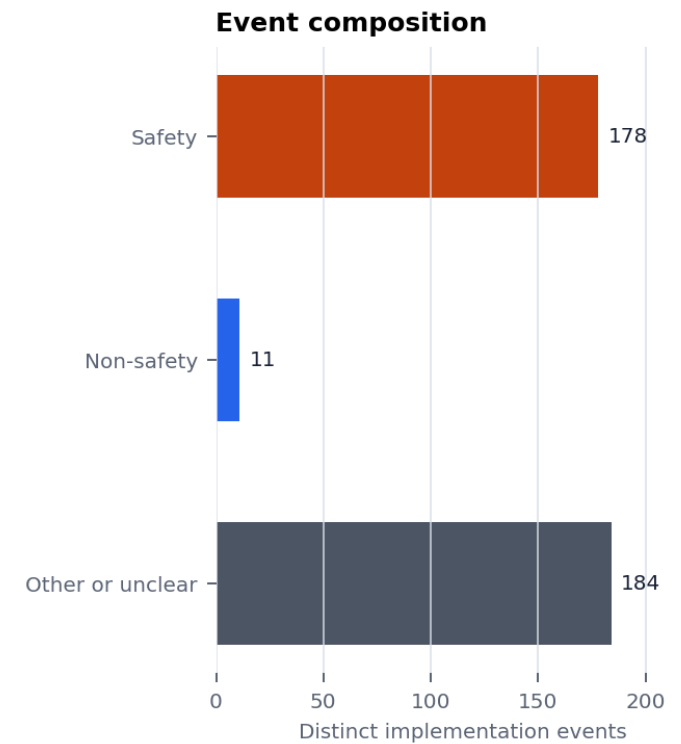
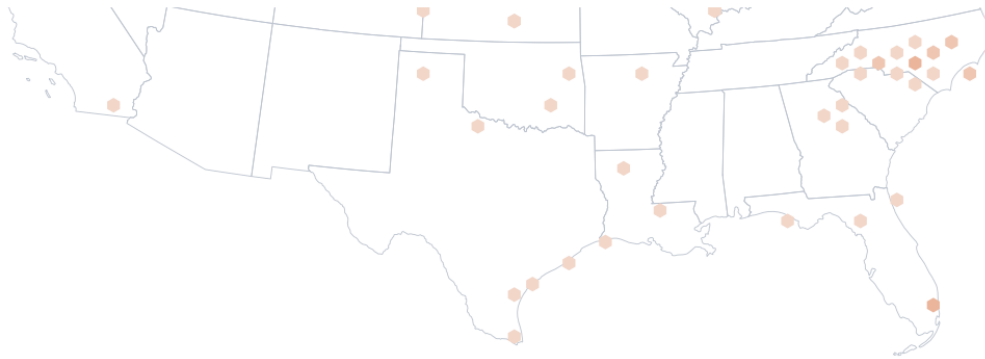
A documented administrative event without a direct pay outcome in the retained evidence.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 544 distinct documented no direct compensation outcome implementation events across 375 municipalities in 28 states. A documented administrative event without a direct pay outcome in the retained evidence. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Across The Board Raise

373 deduplicated implementation events · 298 municipalities · 34 states



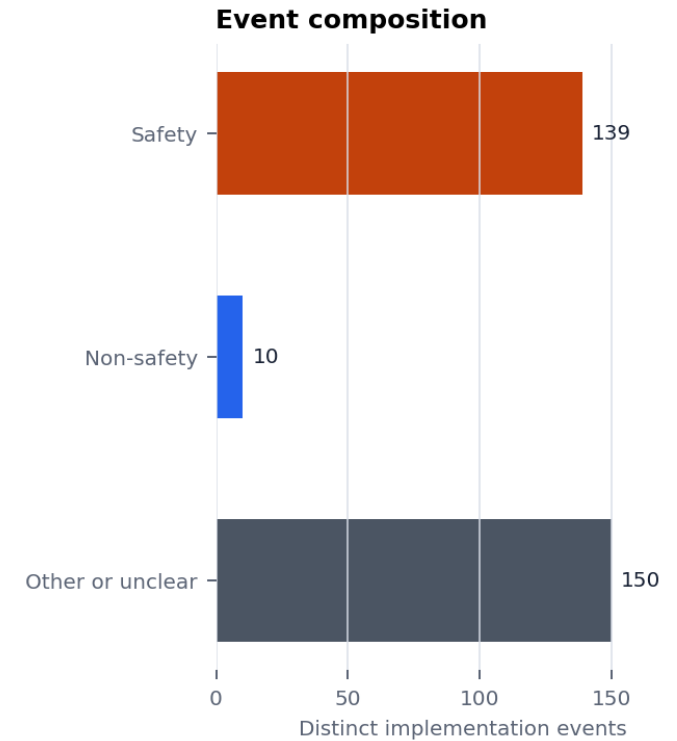
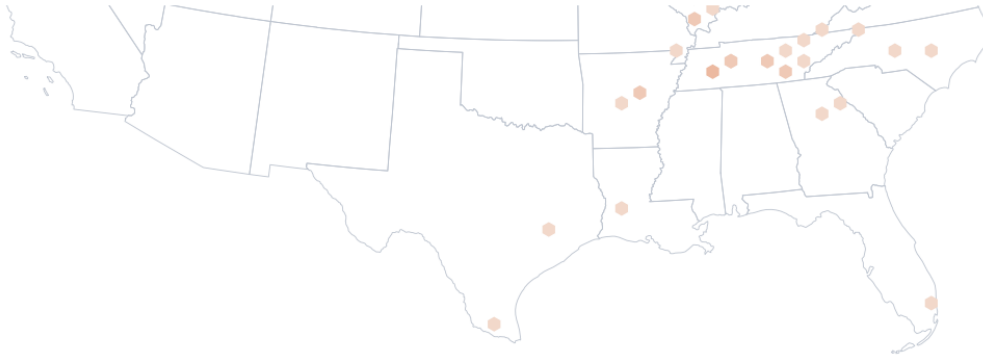
The same general increase applied across covered roles or classifications.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 373 distinct documented across the board raise implementation events across 298 municipalities in 34 states. The same general increase applied across covered roles or classifications. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

One Time

299 deduplicated implementation events · 242 municipalities · 25 states



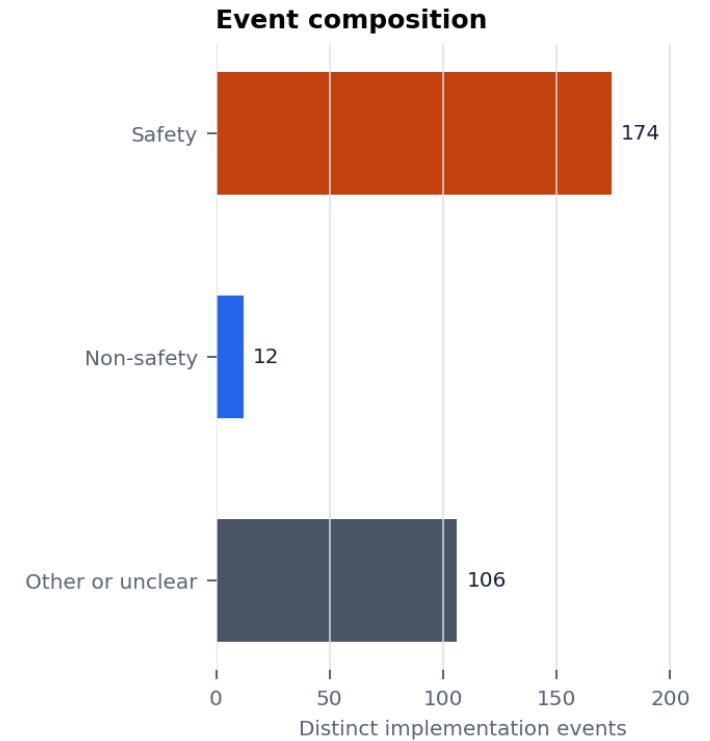
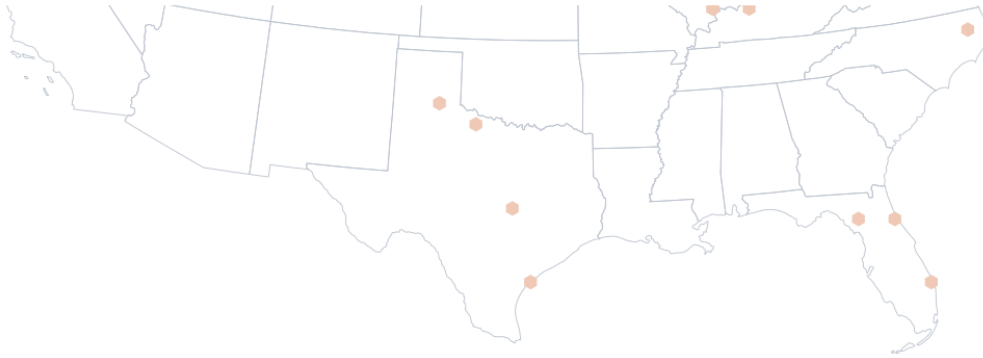
A payment or adjustment that does not recur in the regular base schedule.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 299 distinct documented one time implementation events across 242 municipalities in 25 states. A payment or adjustment that does not recur in the regular base schedule. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Non Base Compensation Other

292 deduplicated implementation events · 195 municipalities · 18 states



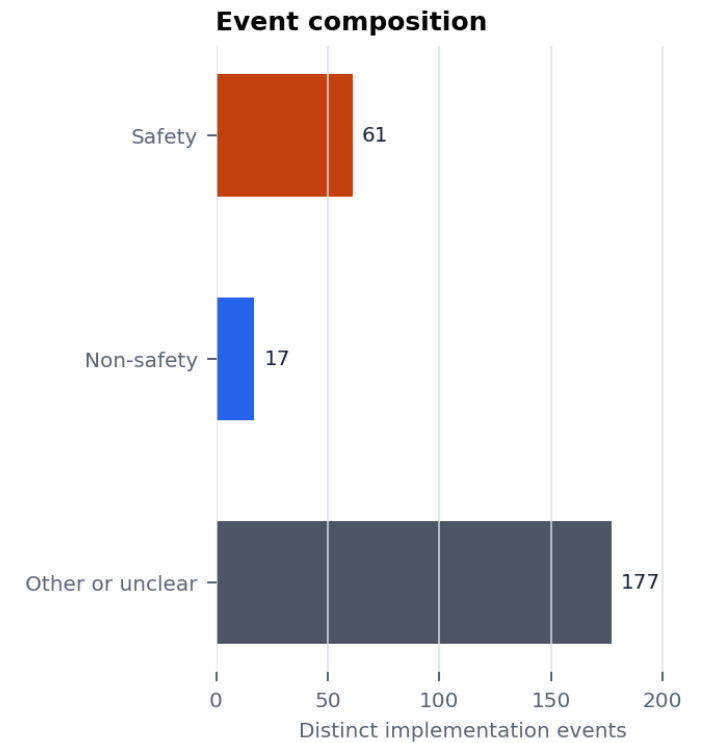
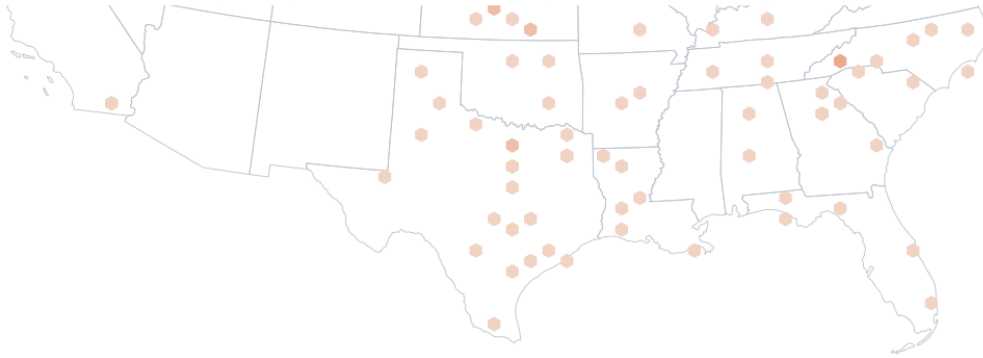
Compensation outside regular base wages that did not fit a more specific category.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 292 distinct documented non base compensation other implementation events across 195 municipalities in 18 states. Compensation outside regular base wages that did not fit a more specific category. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Ordinance Council Adoption

255 deduplicated implementation events · 221 municipalities · 33 states



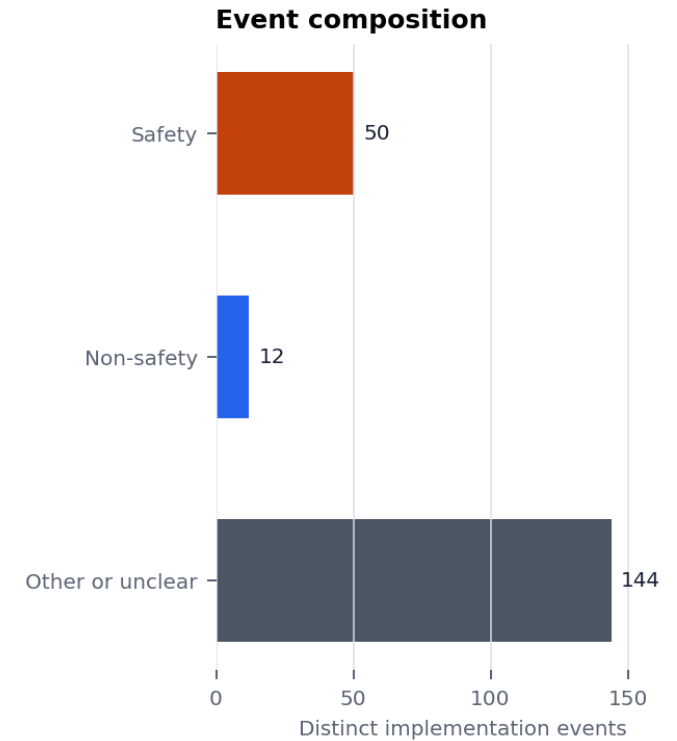
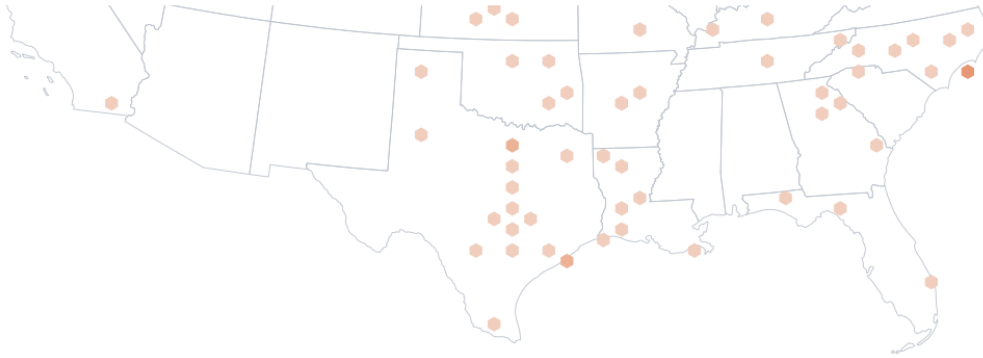
A council or ordinance action formally approving a compensation change.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 255 distinct documented ordinance council adoption implementation events across 221 municipalities in 33 states. A council or ordinance action formally approving a compensation change. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Ordinance Adoption

206 deduplicated implementation events · 188 municipalities · 35 states



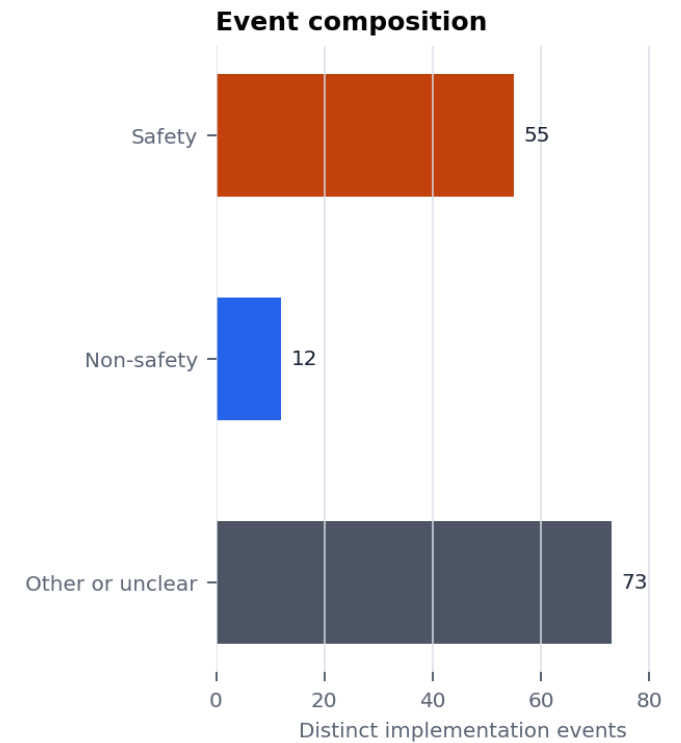
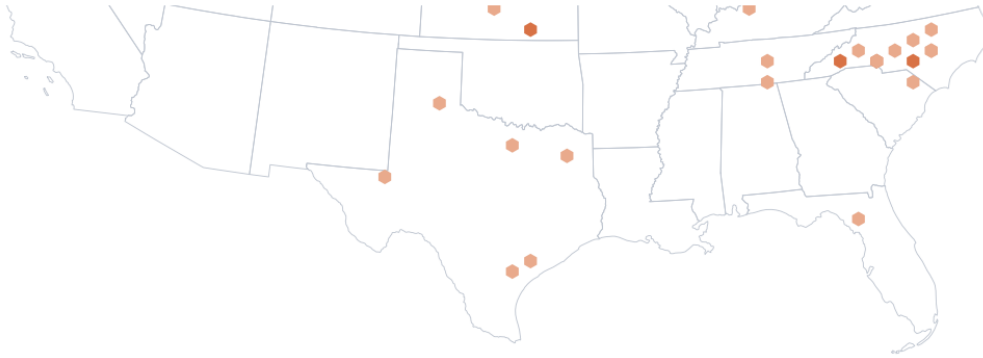
Formal adoption of compensation terms through an ordinance.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 206 distinct documented ordinance adoption implementation events across 188 municipalities in 35 states. Formal adoption of compensation terms through an ordinance. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Unclear

140 deduplicated implementation events · 127 municipalities · 19 states



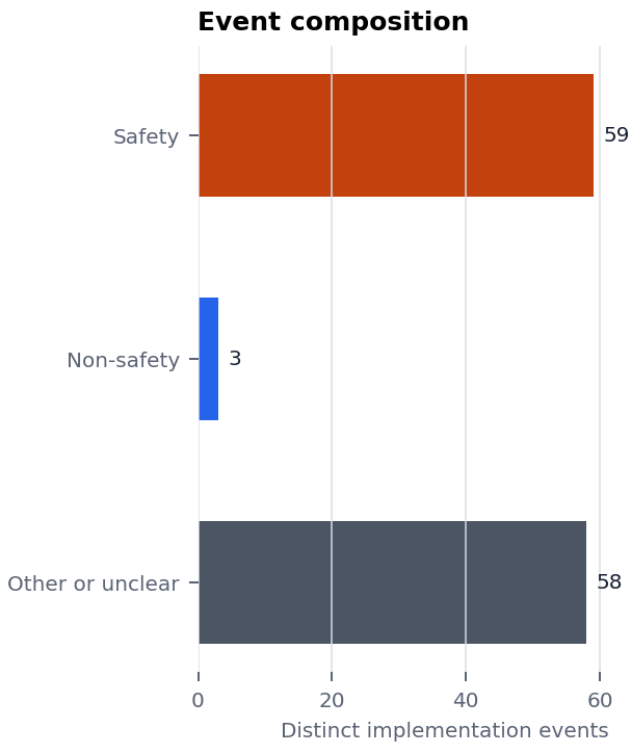
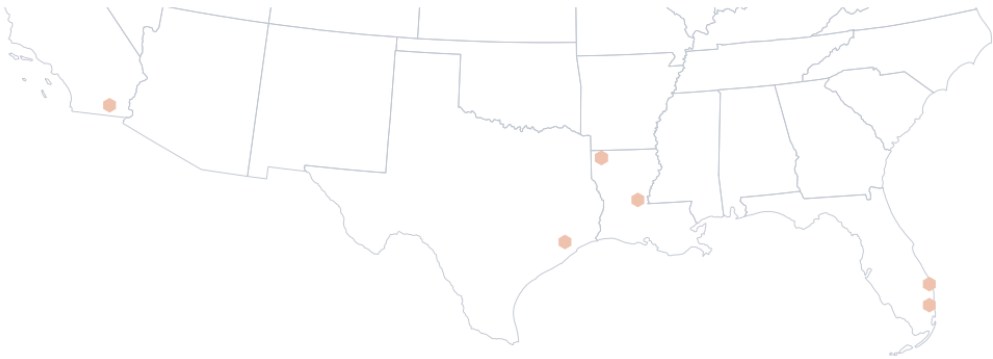
The retained evidence points to a compensation event but does not identify the mechanism clearly.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 140 distinct documented unclear implementation events across 127 municipalities in 19 states. The retained evidence points to a compensation event but does not identify the mechanism clearly. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Base Wage Change

120 deduplicated implementation events · 98 municipalities · 19 states



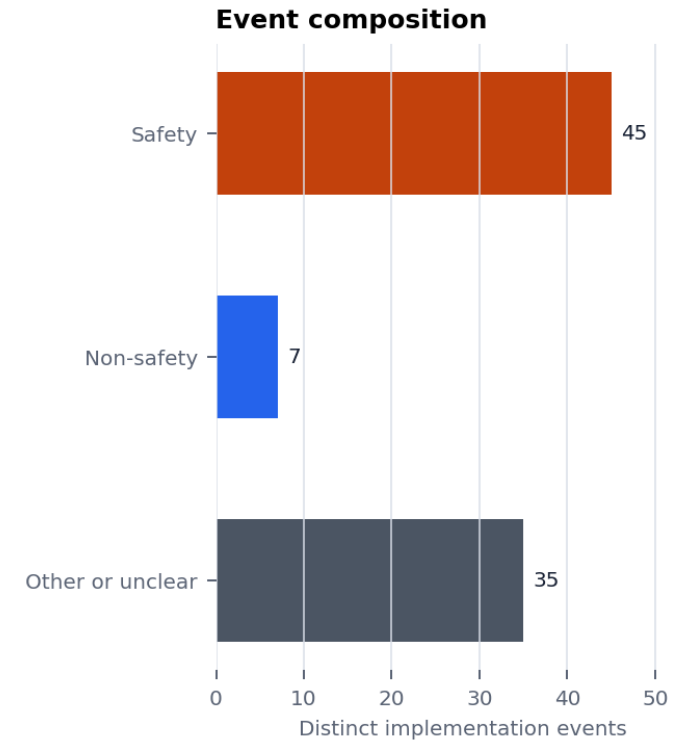
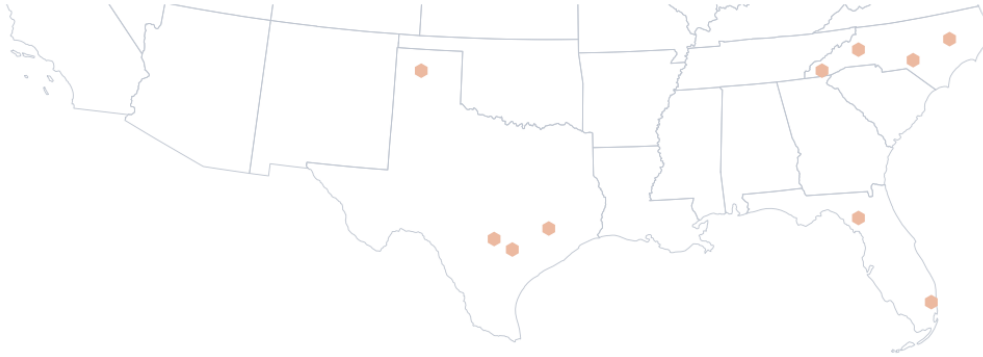
A change to the recurring base wage or salary rate.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 120 distinct documented base wage change implementation events across 98 municipalities in 19 states. A change to the recurring base wage or salary rate. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Step Progression

87 deduplicated implementation events · 74 municipalities · 18 states



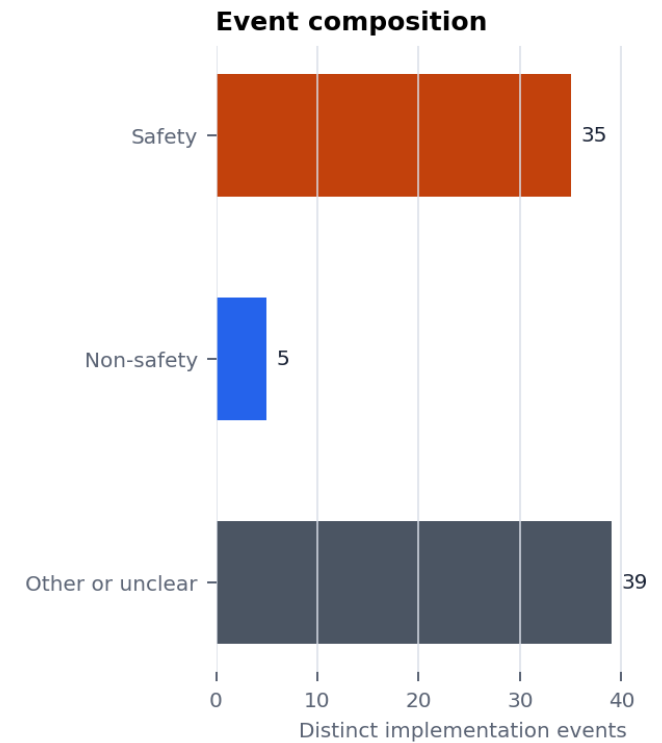
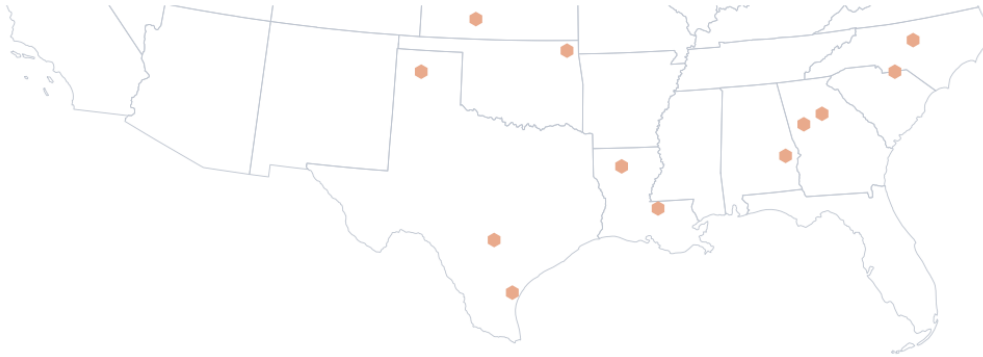
Movement through a salary schedule based on steps, service, rank, or progression rules.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 87 distinct documented step progression implementation events across 74 municipalities in 18 states. Movement through a salary schedule based on steps, service, rank, or progression rules. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

COLA CPI Adjustment

79 deduplicated implementation events · 61 municipalities · 23 states



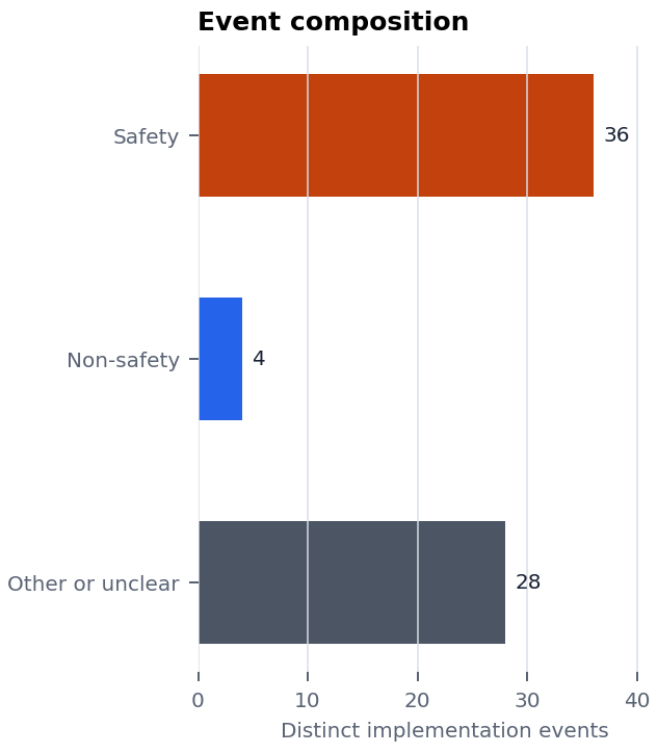
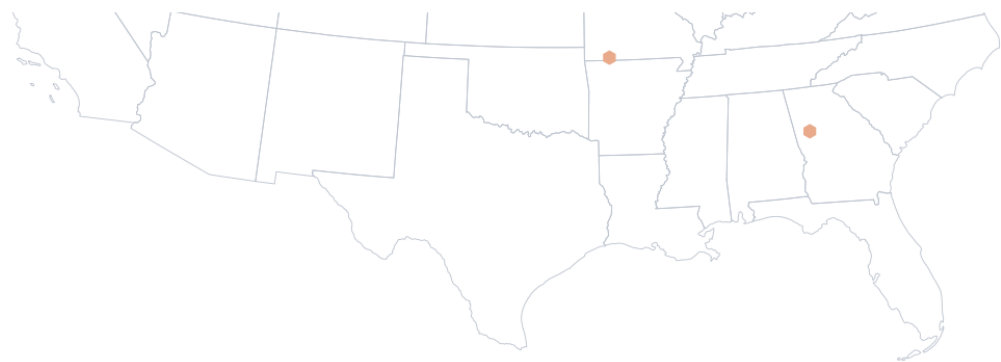
A cost-of-living adjustment, sometimes linked to inflation or a price index.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 79 distinct documented cola cpi adjustment implementation events across 61 municipalities in 23 states. A cost-of-living adjustment, sometimes linked to inflation or a price index. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Uniform Or Equipment Allowance

68 deduplicated implementation events · 56 municipalities · 13 states



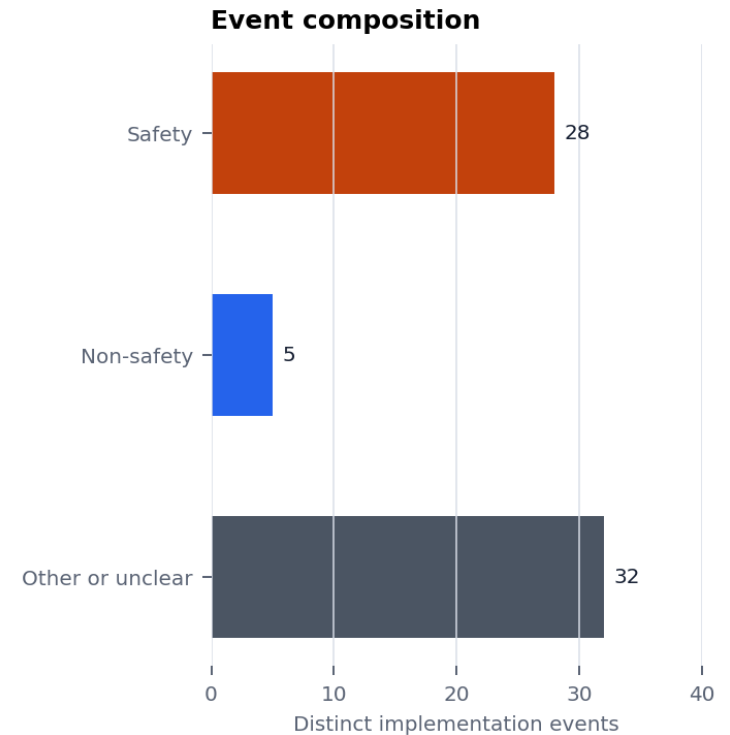
A payment for required uniforms, equipment, maintenance, or related costs.

Analytical unit: municipality x compensation cycle x mechanism x side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 68 distinct documented uniform or equipment allowance implementation events across 56 municipalities in 13 states. A payment for required uniforms, equipment, maintenance, or related costs. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Benefit Cost Change

65 deduplicated implementation events · 53 municipalities · 15 states



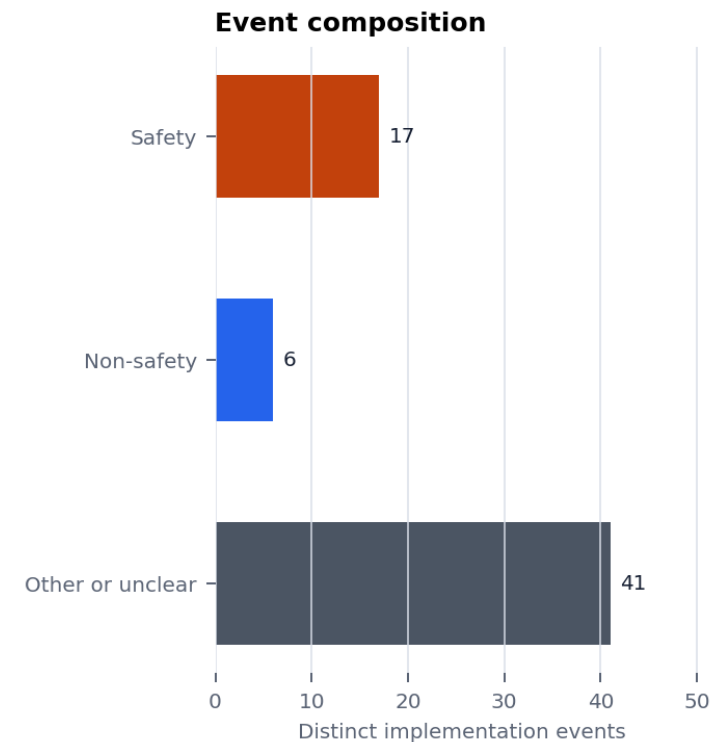
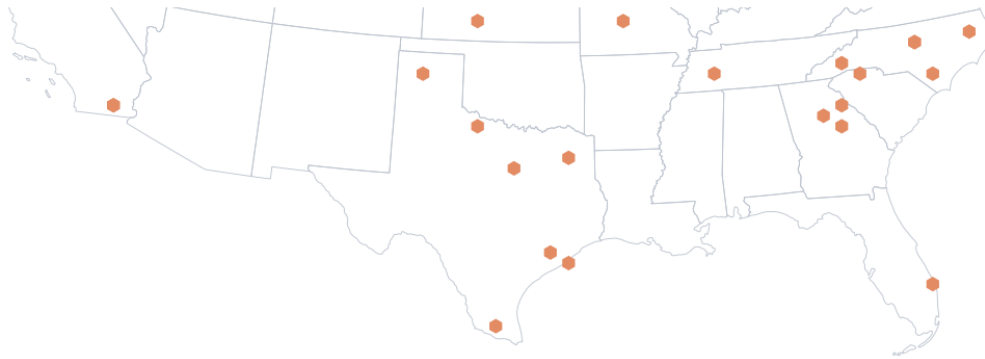
A change in benefit value, employee contributions, or employer benefit costs.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 65 distinct documented benefit cost change implementation events across 53 municipalities in 15 states. A change in benefit value, employee contributions, or employer benefit costs. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Budget Appropriation

64 deduplicated implementation events · 58 municipalities · 24 states



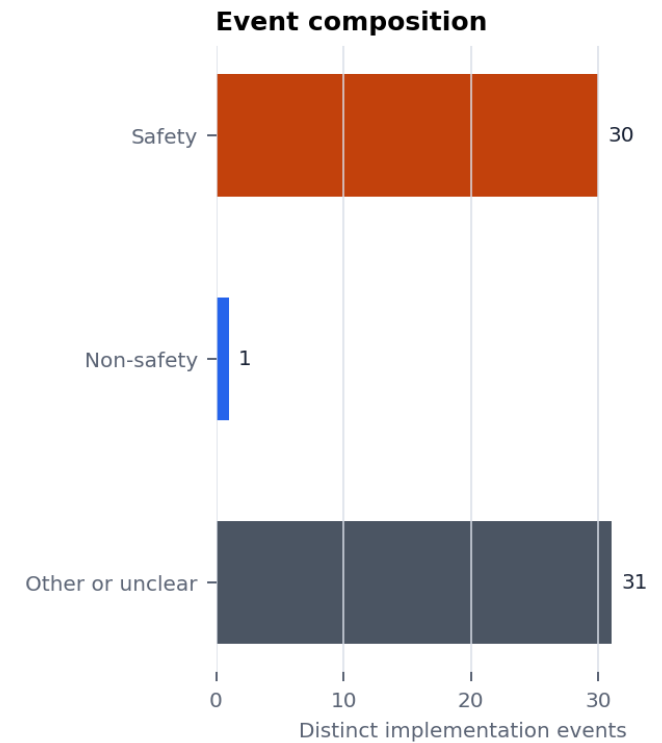
A budget allocation that authorized or funded compensation.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 64 distinct documented budget appropriation implementation events across 58 municipalities in 24 states. A budget allocation that authorized or funded compensation. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Stipend Or Premium

62 deduplicated implementation events · 52 municipalities · 15 states



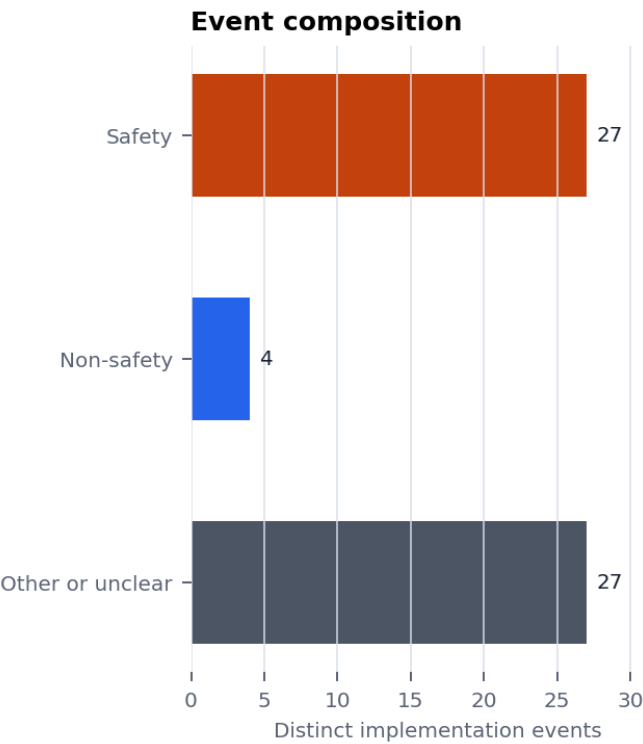
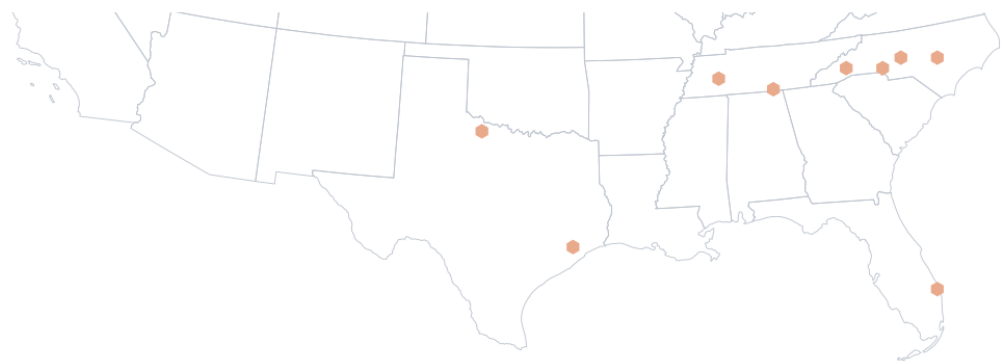
A supplemental payment for assignments, qualifications, shifts, or special duties.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 62 distinct documented stipend or premium implementation events across 52 municipalities in 15 states. A supplemental payment for assignments, qualifications, shifts, or special duties. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Settlement Or MOU

58 deduplicated implementation events · 48 municipalities · 19 states



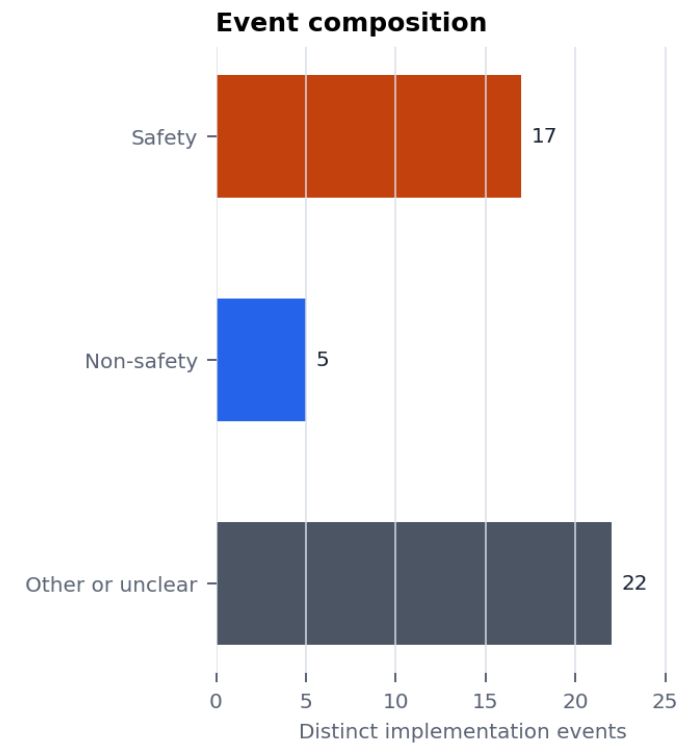
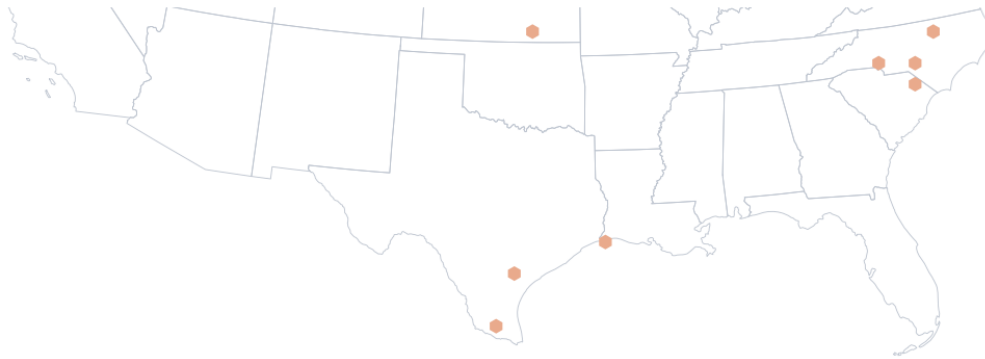
A settlement or memorandum of understanding that changed compensation terms.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 58 distinct documented settlement or mou implementation events across 48 municipalities in 19 states. A settlement or memorandum of understanding that changed compensation terms. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Market Recruitment Retention

44 deduplicated implementation events · 35 municipalities · 16 states



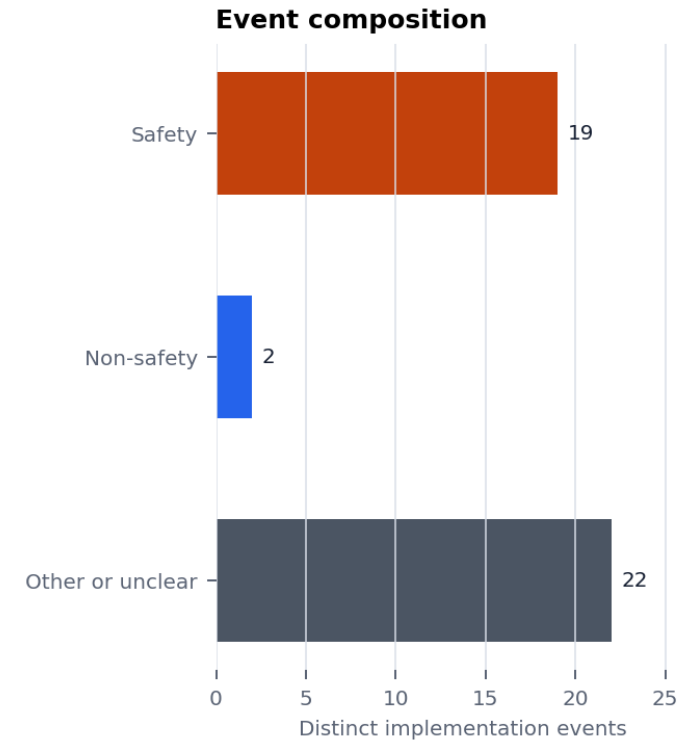
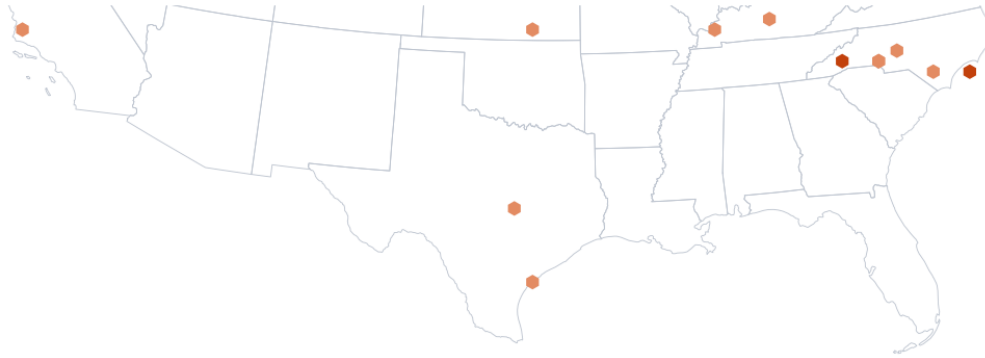
Pay action justified by labor-market competition, hiring, or retention pressure.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 44 distinct documented market recruitment retention implementation events across 35 municipalities in 16 states. Pay action justified by labor-market competition, hiring, or retention pressure. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Classification Band Change

43 deduplicated implementation events · 40 municipalities · 13 states



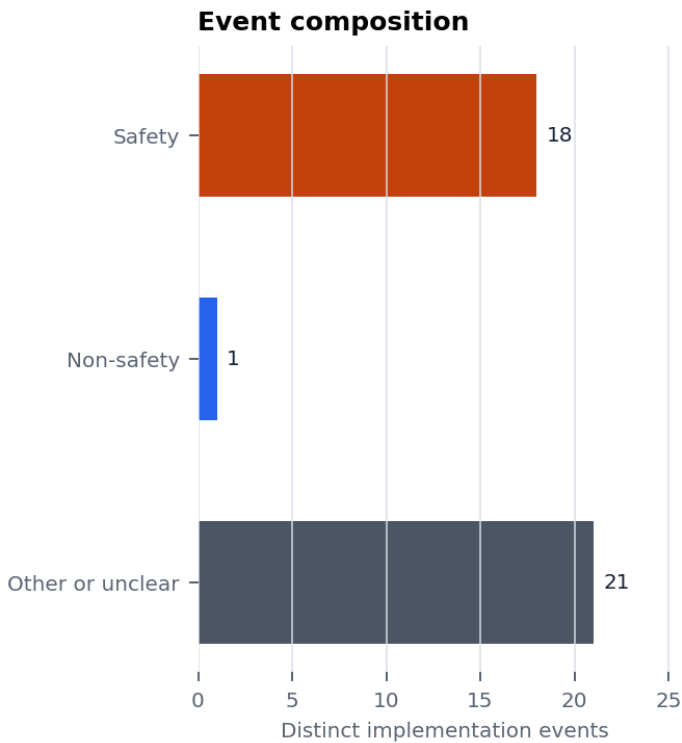
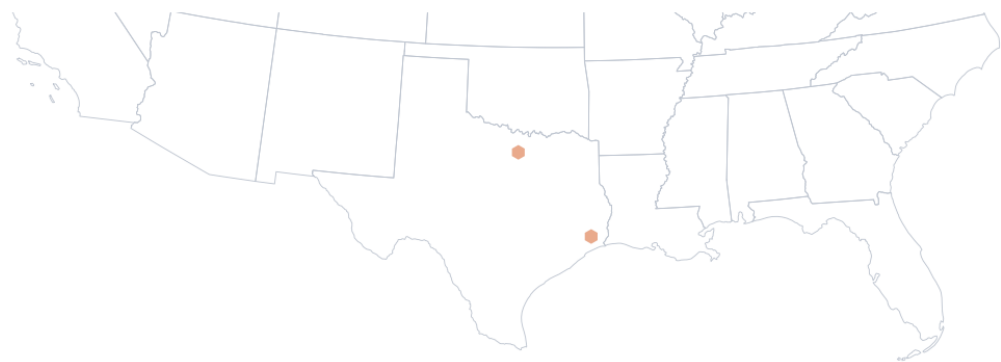
A change to job classification, pay grade, range, or band.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 43 distinct documented classification band change implementation events across 40 municipalities in 13 states. A change to job classification, pay grade, range, or band. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Longevity

40 deduplicated implementation events · 35 municipalities · 13 states



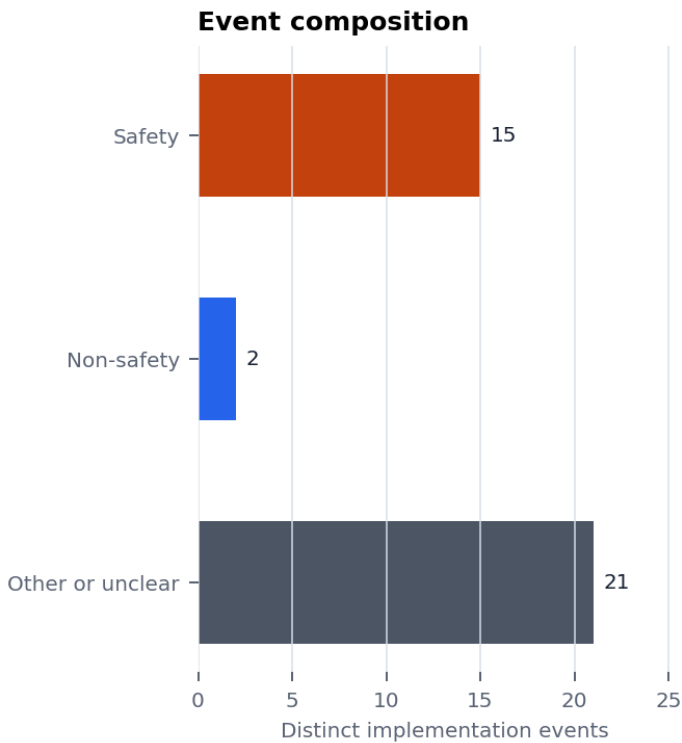
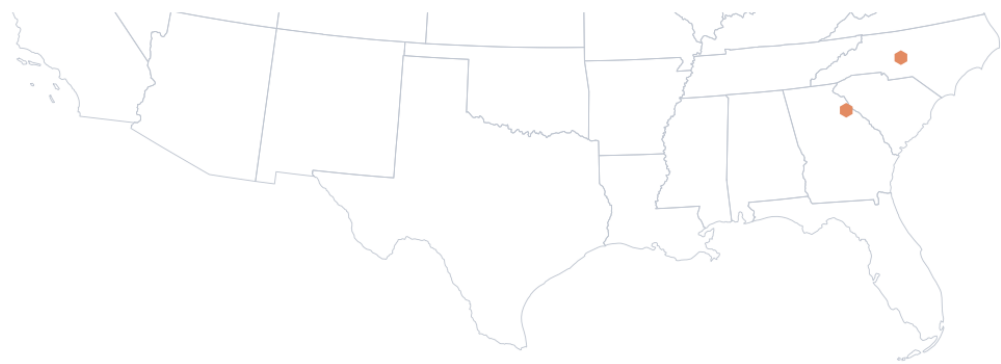
Additional compensation tied to years of service.

Analytical unit: municipality x compensation cycle x mechanism x side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 40 distinct documented longevity implementation events across 35 municipalities in 13 states. Additional compensation tied to years of service. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Overtime

38 deduplicated implementation events · 36 municipalities · 15 states



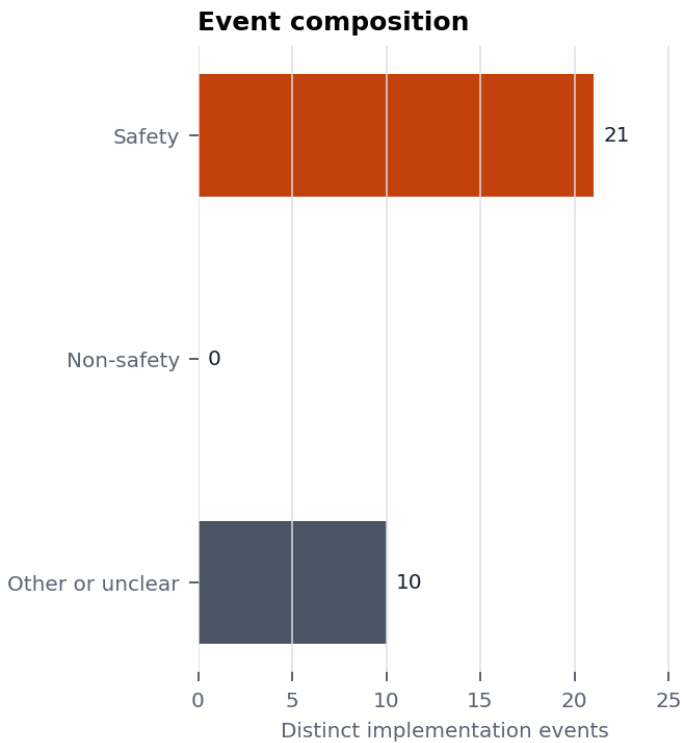
Compensation for hours or duties beyond regular schedules.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 38 distinct documented overtime implementation events across 36 municipalities in 15 states. Compensation for hours or duties beyond regular schedules. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Inflation Indexing

31 deduplicated implementation events · 22 municipalities · 8 states

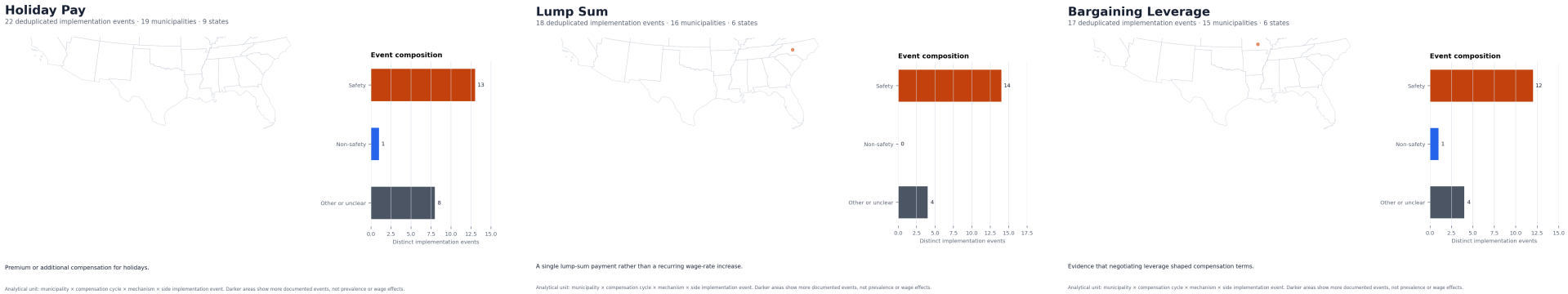


A compensation rule explicitly linked to inflation or an index.

Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

This map shows 31 distinct documented inflation indexing implementation events across 22 municipalities in 8 states. A compensation rule explicitly linked to inflation or an index. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Small-multiple mechanisms



Holiday Pay

This map shows 22 distinct documented holiday pay implementation events across 19 municipalities in 9 states. Premium or additional compensation for holidays. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Lump Sum

This map shows 18 distinct documented lump sum implementation events across 16 municipalities in 6 states. A single lump-sum payment rather than a recurring wage-rate increase. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

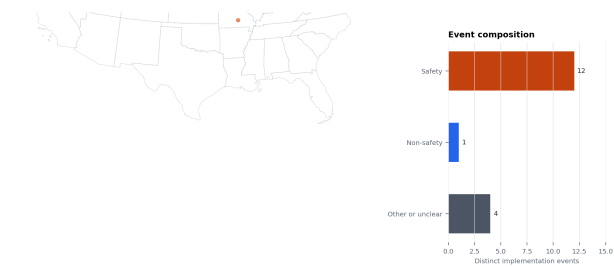
Bargaining Leverage

This map shows 17 distinct documented bargaining leverage implementation events across 15 municipalities in 6 states. Evidence that negotiating leverage shaped compensation terms. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Small-multiple mechanisms

Collective Bargaining

17 deduplicated implementation events - 15 municipalities - 6 states

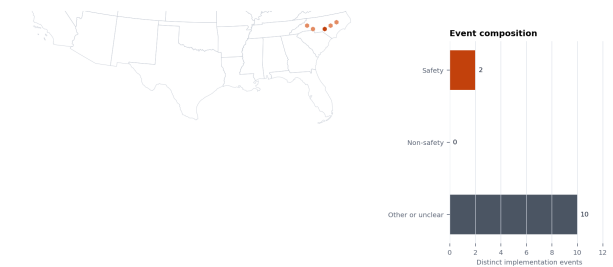


Negotiation between a public employer and represented employees over compensation.

Analytical unit: municipality x compensation cycle x mechanism x side implementation event. Darker areas show more documented events, not prevalence or wage effects.

Salary Range Change

12 deduplicated implementation events - 12 municipalities - 5 states



A change to the minimum, maximum, or span of a salary range.

Analytical unit: municipality x compensation cycle x mechanism x side implementation event. Darker areas show more documented events, not prevalence or wage effects.

Contract Ratification

9 deduplicated implementation events - 9 municipalities - 6 states



Formal approval of negotiated contract terms by the relevant parties.

Analytical unit: municipality x compensation cycle x mechanism x side implementation event. Darker areas show more documented events, not prevalence or wage effects.

Collective Bargaining

This map shows 17 distinct documented collective bargaining implementation events across 15 municipalities in 6 states. Negotiation between a public employer and represented employees over compensation. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Salary Range Change

This map shows 12 distinct documented salary range change implementation events across 12 municipalities in 5 states. A change to the minimum, maximum, or span of a salary range. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

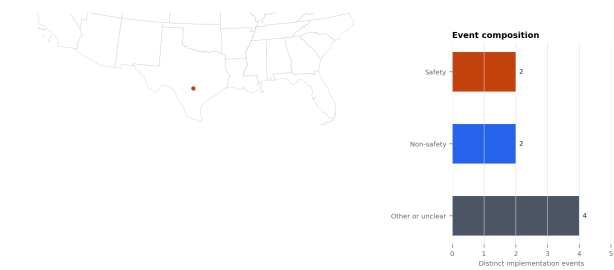
Contract Ratification

This map shows 9 distinct documented contract ratification implementation events across 9 municipalities in 6 states. Formal approval of negotiated contract terms by the relevant parties. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Small-multiple mechanisms

Vacancy Pressure

8 deduplicated implementation events · 8 municipalities · 4 states



Unfilled positions or vacancy conditions cited in compensation or staffing decisions.
Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

Reimbursement

7 deduplicated implementation events · 6 municipalities · 4 states



Repayment of eligible work-related expenses rather than ordinary wages.
Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

Comparability Parity

6 deduplicated implementation events · 6 municipalities · 5 states



Compensation linked to comparison with another role, unit, or jurisdiction.
Analytical unit: municipality × compensation cycle × mechanism × side implementation event. Darker areas show more documented events, not prevalence or wage effects.

Vacancy Pressure

This map shows 8 distinct documented vacancy pressure implementation events across 8 municipalities in 4 states. Unfilled positions or vacancy conditions cited in compensation or staffing decisions. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Reimbursement

This map shows 7 distinct documented reimbursement implementation events across 6 municipalities in 4 states. Repayment of eligible work-related expenses rather than ordinary wages. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Comparability Parity

This map shows 6 distinct documented comparability parity implementation events across 6 municipalities in 5 states. Compensation linked to comparison with another role, unit, or jurisdiction. The geography matters because it locates the process in the retained evidence. Darker areas contain more events, not more workers or larger wage effects, and the map is not a national prevalence estimate.

Sparse mechanisms

Classification Civil Service

4 deduplicated implementation events · 4 municipalities · 2 states



Sparse evidence

- Marion, OH**
Fire · Formally Adopted
- Perry, OH**
Police · Implemented
- Put-in-Bay, OH**
Remains Unclear · Formally Adopted
- Holland, TX**
Remains Unclear · Formally Adopted

Civil-service or classification rules that structure pay placement or progression.

Analytical unit: municipality x compensation cycle x mechanism x side implementation event. Darker areas show more documented events, not prevalence or wage effects.

Interest Arbitration

1 deduplicated implementation events · 1 municipalities · 1 states



Sparse evidence

- Jefferson, WI**
Police · Formally Adopted

A neutral decision process that resolves bargaining impasses over contract terms.

Analytical unit: municipality x compensation cycle x mechanism x side implementation event. Darker areas show more documented events, not prevalence or wage effects.

Classification Civil Service

I found 4 documented classification civil service implementation events across 4 municipalities in 2 states. Civil-service or classification rules that structure pay placement or progression. The evidence is too sparse for a density pattern, so I show the exact locations instead. This establishes that the mechanism appears in the retained evidence; it does not show how common it is or how much it changed wages.

Interest Arbitration

I found 1 documented interest arbitration implementation event across 1 municipality in 1 state. A neutral decision process that resolves bargaining impasses over contract terms. The evidence is too sparse for a density pattern, so I show the exact locations instead. This establishes that the mechanism appears in the retained evidence; it does not show how common it is or how much it changed wages.

Mechanism glossary

None Identified (3,694)	No specific compensation mechanism was identified in the retained event record.
Payroll Effective Date (1,876)	The date a compensation change became operative in payroll administration.
Retroactive (1,418)	A compensation change applied to an earlier effective period.
Retroactive Pay (1,418)	Back pay issued after a compensation term took effect or was settled.
Recurring (720)	A compensation component that continues rather than ending after one payment.
Fiscal Constraint (619)	Budget pressure or limited fiscal capacity that constrained compensation choices.
Budget Pay Plan Process (617)	A budget or pay-plan process that formally set or revised compensation.
No Direct Compensation Outcome (544)	A documented administrative event without a direct pay outcome in the retained evidence.
Across The Board Raise (373)	The same general increase applied across covered roles or classifications.
One Time (299)	A payment or adjustment that does not recur in the regular base schedule.
Non Base Compensation Other (292)	Compensation outside regular base wages that did not fit a more specific category.
Ordinance Council Adoption (255)	A council or ordinance action formally approving a compensation change.
Ordinance Adoption (206)	Formal adoption of compensation terms through an ordinance.
Unclear (140)	The retained evidence points to a compensation event but does not identify the mechanism clearly.
Base Wage Change (120)	A change to the recurring base wage or salary rate.
Step Progression (87)	Movement through a salary schedule based on steps, service, rank, or progression rules.
COLA CPI Adjustment (79)	A cost-of-living adjustment, sometimes linked to inflation or a price index.
Uniform Or Equipment Allowance (68)	A payment for required uniforms, equipment, maintenance, or related costs.
Benefit Cost Change (65)	A change in benefit value, employee contributions, or employer benefit costs.

Mechanism glossary (continued)

Budget Appropriation (64)	A budget allocation that authorized or funded compensation.
Stipend Or Premium (62)	A supplemental payment for assignments, qualifications, shifts, or special duties.
Settlement Or MOU (58)	A settlement or memorandum of understanding that changed compensation terms.
Market Recruitment Retention (44)	Pay action justified by labor-market competition, hiring, or retention pressure.
Classification Band Change (43)	A change to job classification, pay grade, range, or band.
Longevity (40)	Additional compensation tied to years of service.
Overtime (38)	Compensation for hours or duties beyond regular schedules.
Inflation Indexing (31)	A compensation rule explicitly linked to inflation or an index.
Holiday Pay (22)	Premium or additional compensation for holidays.
Lump Sum (18)	A single lump-sum payment rather than a recurring wage-rate increase.
Bargaining Leverage (17)	Evidence that negotiating leverage shaped compensation terms.
Collective Bargaining (17)	Negotiation between a public employer and represented employees over compensation.
Salary Range Change (12)	A change to the minimum, maximum, or span of a salary range.
Contract Ratification (9)	Formal approval of negotiated contract terms by the relevant parties.
Vacancy Pressure (8)	Unfilled positions or vacancy conditions cited in compensation or staffing decisions.
Reimbursement (7)	Repayment of eligible work-related expenses rather than ordinary wages.
Comparability Parity (6)	Compensation linked to comparison with another role, unit, or jurisdiction.
Classification Civil Service (4)	Civil-service or classification rules that structure pay placement or progression.
Interest Arbitration (1)	A neutral decision process that resolves bargaining impasses over contract terms.

Mechanism glossary (continued)

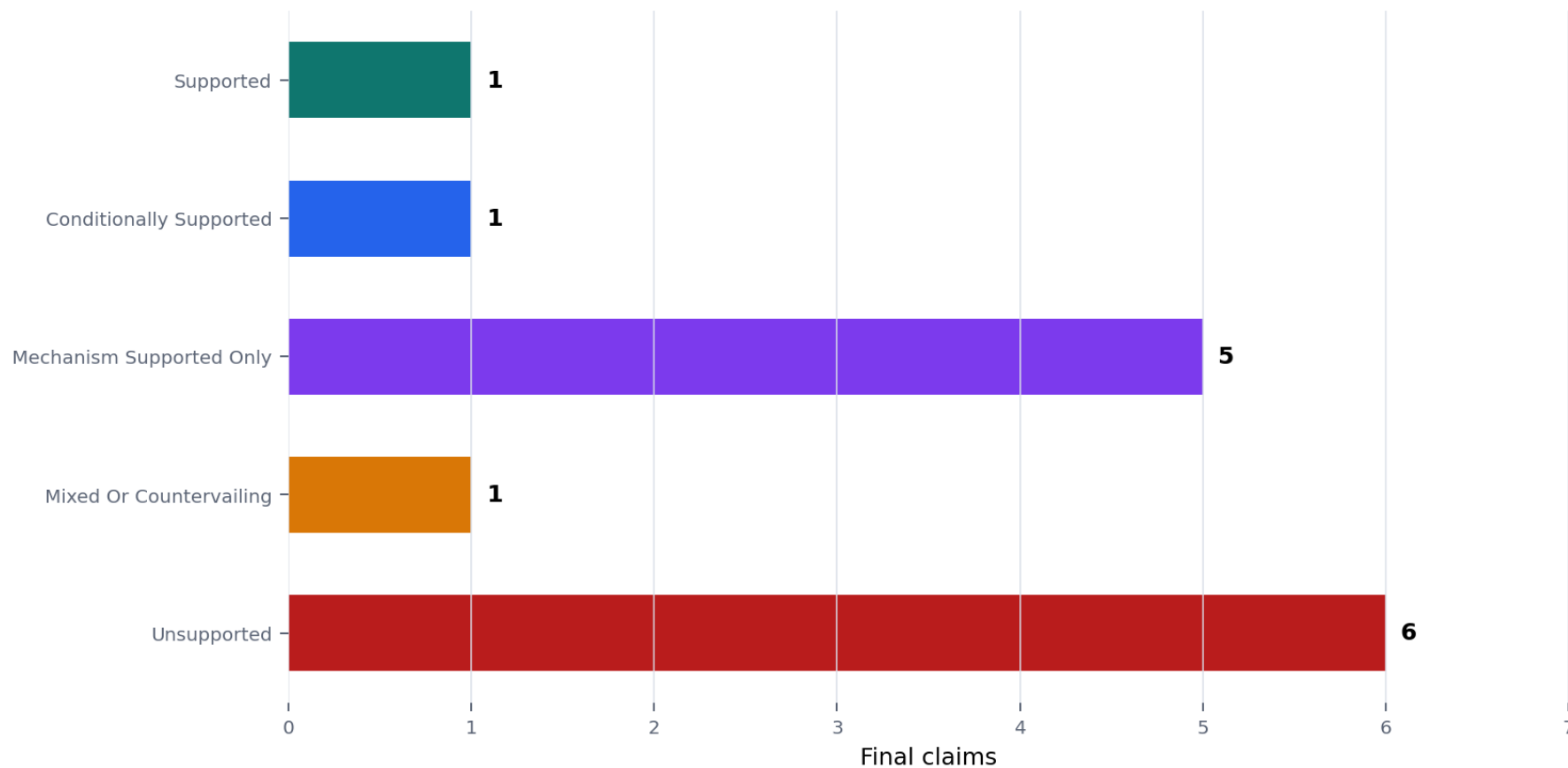
Section II

What the claims say

A visual section built from adjudicated, source-linked evidence.

What the final claims support

Fourteen claims, each assigned one evidence-bounded class



A mechanism-supported claim shows how a process operates; it does not establish an average wage effect, prevalence, or causality.

This overview preserves the final analytical unit and evidence boundary. Counts show documented evidence in this project, not national prevalence or causal effects.

Claim evidence matrix

Evidence counts are capped visually at 20; printed values show the exact bounded counts

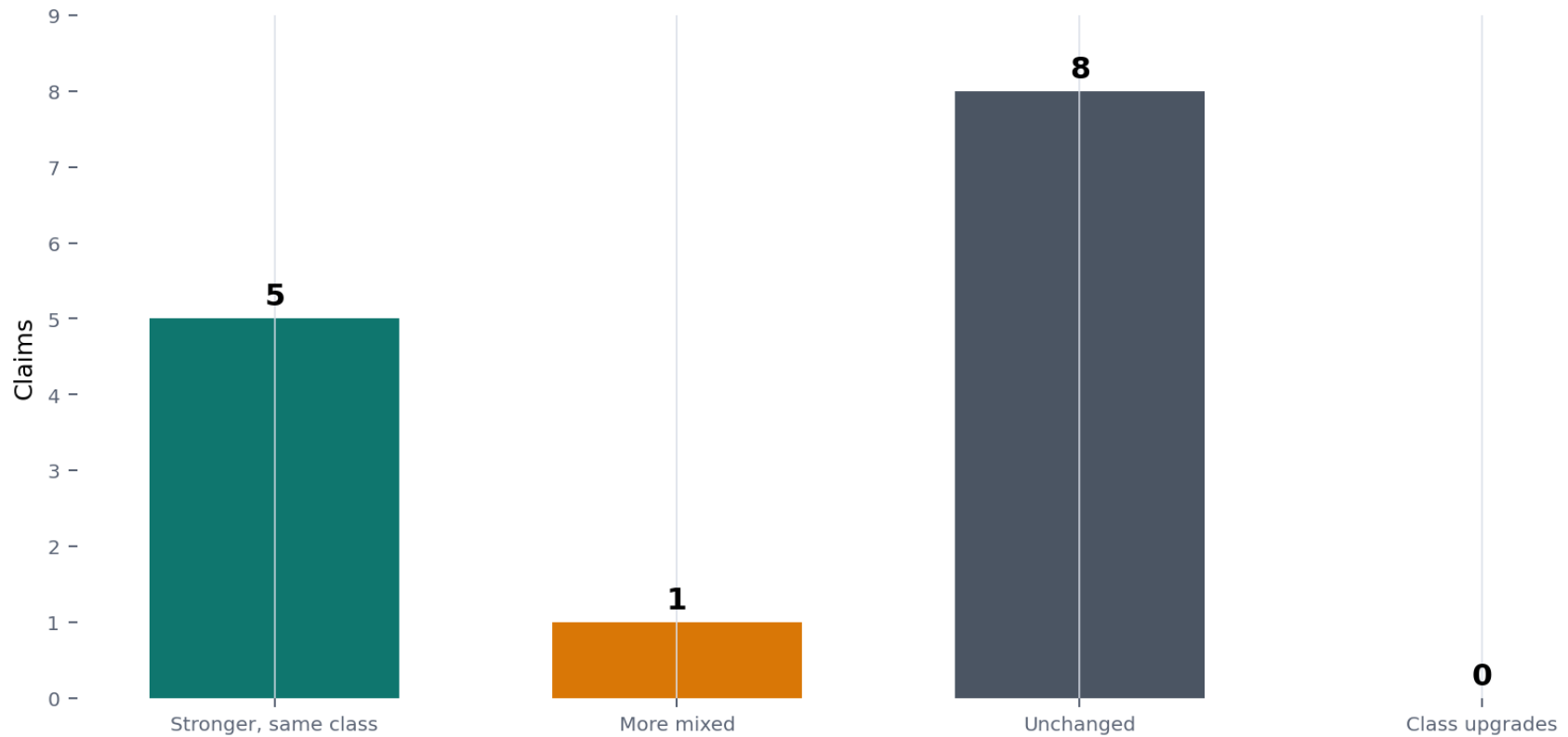
Formal bargaining institutions shape compens	6	0	1	0	0
Scheduled growth mechanisms can differ by oc	7	6	20	0	0
Non-base pay creates additional compensation	2	0	1	0	0
Staffing pressure can enter compensation dec	4	7	20	0	0
Retroactivity can turn delay into back pay	20	0	1	0	0
Budgets and ordinances formalize or constrai	20	0	1	0	0
The safety and non-safety pattern is mixed	11	15	20	7	0
The evidence supports a bounded mechanism ac	20	20	20	7	0
A national safety wage gap is not establishe	0	0	0	0	0
National mechanism prevalence is not establi	0	0	0	0	0
A causal effect is not established	0	0	0	0	0
A regression result is not available	0	0	0	0	0
A fixed safety growth advantage is not estab	0	0	0	0	0
Mechanisms do not prove a wage difference	0	0	0	0	0
	Strict	Bounded	Directional	Counterexamples	Conflicts

Rows are claims, not observations. Strict, bounded, and directional evidence remain distinct; unresolved conflicts are shown but excluded from support.

This overview preserves the final analytical unit and evidence boundary. Counts show documented evidence in this project, not national prevalence or causal effects.

Broader evidence changed depth, not the final claim classes

Five claims strengthened within their existing class; one became more mixed; none moved to a stronger class

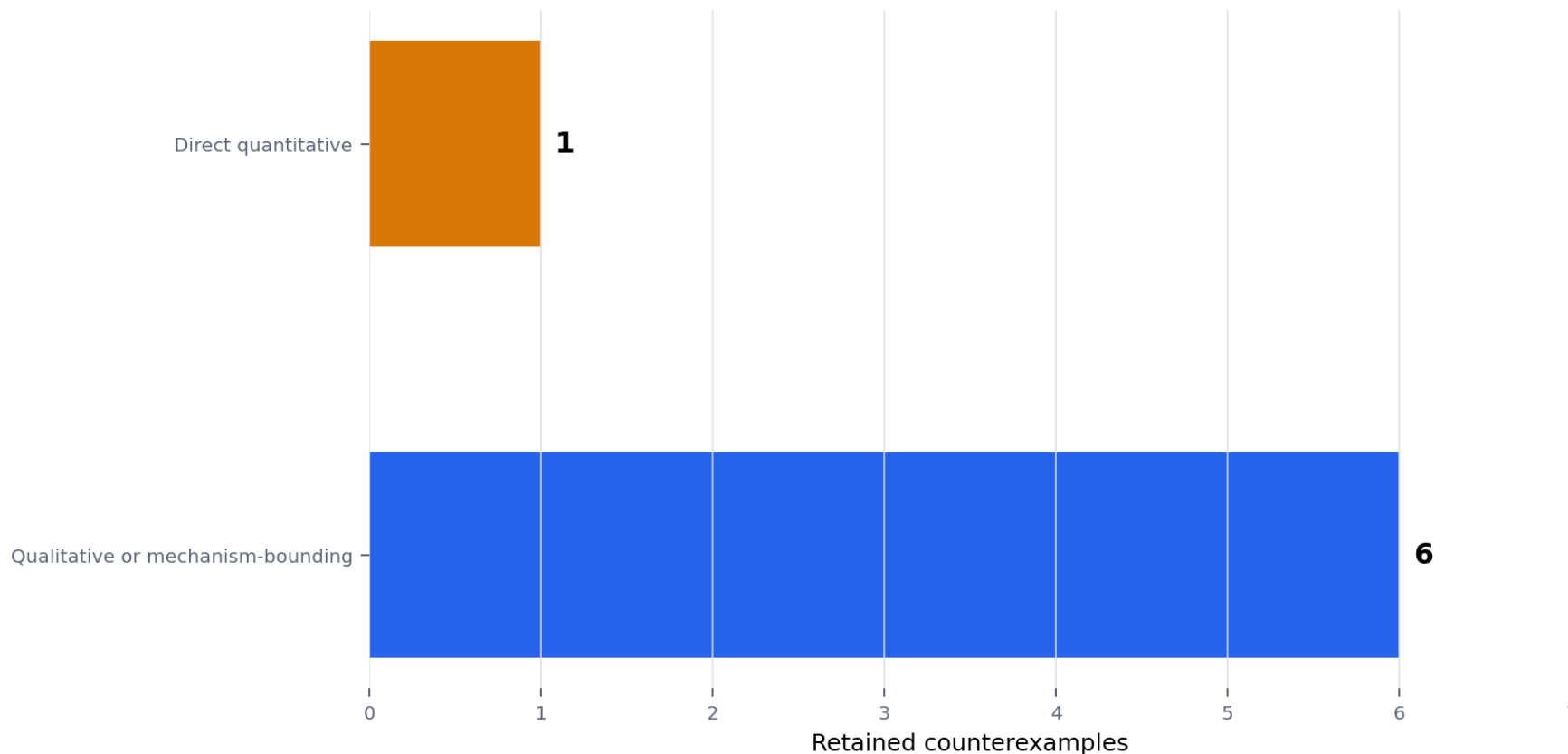


The broader analysis admitted bounded and directional evidence without erasing the strict baseline or upgrading unsupported national and causal claims.

This overview preserves the final analytical unit and evidence boundary. Counts show documented evidence in this project, not national prevalence or causal effects.

Counterexamples remain part of the result

One local comparison directly runs against a simple safety-wage advantage; six other records bound mechanisms or generalizations



Counterexamples are not discarded as noise. They define the edge of defensible wording and keep local or mechanism-specific patterns from becoming national claims.

Seven retained counterexamples: one direct quantitative and six qualitative, conditional, or mechanism-bounding records.

This overview preserves the final analytical unit and evidence boundary. Counts show documented evidence in this project, not national prevalence or causal effects.

Formal bargaining institutions shape compensation terms

CLAIM-A · Mechanism Supported Only

Strict conclusion

In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms.

Broader bounded conclusion

In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms; the evidence establishes mechanism operation, not an average wage effect.

What I can responsibly say

In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms; the evidence establishes mechanism operation, not an average wage effect.

What I cannot responsibly say

Formal bargaining causes safety wages to rise faster nationwide.

Strict evidence 6 · Bounded 0 · Directional 1 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Supporting Finding. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as mechanism supported only. In the reviewed corpus, formal bargaining, arbitration, and factfinding provide recurring institutional channels that can set or preserve safety-compensation terms. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

Scheduled growth mechanisms can differ by occupation

CLAIM-B · Conditionally Supported

Strict conclusion

The reviewed documentary corpus contains recurring step, seniority, rank, and COLA structures; comparative growth implications remain sample-specific.

Broader bounded conclusion

In reviewed unit-cycle records, step, seniority, rank, and COLA provisions create recurring scheduled wage-growth paths; step progression leans safety in the available sample, while across-board results are mixed and COLA cells are sparse.

What I can responsibly say

In reviewed unit-cycle records, step, seniority, rank, and COLA provisions create recurring scheduled wage-growth paths; step progression leans safety in the available sample, while across-board results are mixed and COLA cells are sparse.

What I cannot responsibly say

Safety wages generally grow faster nationwide because of step progression or COLAs.

Strict evidence 7 · Bounded 6 · Directional 423 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Major Section. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as conditionally supported. The reviewed documentary corpus contains recurring step, seniority, rank, and COLA structures; comparative growth implications remain sample-specific. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

Non-base pay creates additional compensation channels

CLAIM-C · Mechanism Supported Only

Strict conclusion

Reviewed safety-compensation records contain structured non-base channels—including overtime, holiday, longevity, premium, stipend, and allowance provisions—that can raise compensation beyond base pay.

Broader bounded conclusion

Reviewed safety-compensation records contain structured non-base channels—including overtime, holiday, longevity, premium, stipend, and allowance provisions—that can raise compensation beyond base pay; no complete total-compensation sum is estimated.

What I can responsibly say

Reviewed safety-compensation records contain structured non-base channels—including overtime, holiday, longevity, premium, stipend, and allowance provisions—that can raise compensation beyond base pay; no complete total-compensation sum is estimated.

What I cannot responsibly say

Safety workers receive more total compensation than non-safety workers nationwide.

Strict evidence 2 · Bounded 0 · Directional 1 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Supporting Finding. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as mechanism supported only. Reviewed safety-compensation records contain structured non-base channels—including overtime, holiday, longevity, premium, stipend, and allowance provisions—that can raise compensation beyond base pay. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

Staffing pressure can enter compensation decisions

CLAIM-D · Mechanism Supported Only

Strict conclusion

Reviewed sources explicitly use recruitment, retention, comparator, vacancy, and staffing pressure to justify schedule changes or targeted premiums in some public-safety settings.

Broader bounded conclusion

Reviewed sources explicitly use recruitment, retention, comparator, vacancy, and staffing pressure to justify schedule changes or targeted premiums in some public-safety settings; the staffing evidence is descriptive, not causal or prevalence evidence.

What I can responsibly say

Reviewed sources explicitly use recruitment, retention, comparator, vacancy, and staffing pressure to justify schedule changes or targeted premiums in some public-safety settings; the staffing evidence is descriptive, not causal or prevalence evidence.

What I cannot responsibly say

Staffing shortages cause higher safety wages.

Strict evidence 4 · Bounded 7 · Directional 217 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Supporting Finding. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as mechanism supported only. Reviewed sources explicitly use recruitment, retention, comparator, vacancy, and staffing pressure to justify schedule changes or targeted premiums in some public-safety settings. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

Retroactivity can turn delay into back pay

CLAIM-E · Mechanism Supported Only

Strict conclusion

Reviewed records show that retroactive effective dates and implementation clauses can convert some bargaining delays into payable increases or back pay.

Broader bounded conclusion

Reviewed records show that retroactive effective dates and implementation clauses can convert some bargaining delays into payable increases or back pay; adoption alone is not payment, and only retained evidence with a paid stage is described as paid.

What I can responsibly say

Reviewed records show that retroactive effective dates and implementation clauses can convert some bargaining delays into payable increases or back pay; adoption alone is not payment, and only retained evidence with a paid stage is described as paid.

What I cannot responsibly say

Every adopted or retroactive provision was implemented and paid.

Strict evidence 40 · Bounded 0 · Directional 1 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Supporting Finding. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as mechanism supported only. Reviewed records show that retroactive effective dates and implementation clauses can convert some bargaining delays into payable increases or back pay. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

Budgets and ordinances formalize or constrain pay

CLAIM-F · Mechanism Supported Only

Strict conclusion

Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions.

Broader bounded conclusion

Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions; these records do not establish that every proposal was implemented or paid.

What I can responsibly say

Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions; these records do not establish that every proposal was implemented or paid.

What I cannot responsibly say

Budget and council action uniformly increase compensation.

Strict evidence 44 · Bounded 0 · Directional 1 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Supporting Finding. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as mechanism supported only. Reviewed ordinances, budgets, pay plans, and classification records show how compensation proposals can be formalized or constrained by fiscal institutions. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

The safety and non-safety pattern is mixed

CLAIM-G · Mixed Or Countervailing

Strict conclusion

Safety and non-safety units both use formal compensation-growth mechanisms; the comparative pattern in reviewed records is mixed.

Broader bounded conclusion

Both safety and non-safety units use bargaining, steps, COLAs, and pay plans. The available evidence is consistent with a reinforcing safety-pressure bundle in some settings, but local and growth results are mixed and do not establish a uniform safety advantage.

What I can responsibly say

Both safety and non-safety units use bargaining, steps, COLAs, and pay plans. The available evidence is consistent with a reinforcing safety-pressure bundle in some settings, but local and growth results are mixed and do not establish a uniform safety advantage.

What I cannot responsibly say

Non-safety compensation lacks institutional growth mechanisms, or safety always has the advantage.

Strict evidence 11 · Bounded 15 · Directional 424 · Counterexamples 7 · Conflicts 0

Recommended placement: Report Counterexample Section. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as mixed or countervailing. Safety and non-safety units both use formal compensation-growth mechanisms; the comparative pattern in reviewed records is mixed. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

The evidence supports a bounded mechanism account

CLAIM-H · Supported

Strict conclusion

The reviewed corpus supports a bounded account of mechanisms and directional pressures affecting safety compensation more strongly than it supports any global wage-gap, prevalence, or causal-effect estimate.

Broader bounded conclusion

The reviewed corpus supports a bounded account of mechanisms and directional pressures affecting safety compensation more strongly than it supports any global wage-gap, prevalence, or causal-effect estimate.

What I can responsibly say

The reviewed corpus supports a bounded account of mechanisms and directional pressures affecting safety compensation more strongly than it supports any global wage-gap, prevalence, or causal-effect estimate.

What I cannot responsibly say

The corpus estimates a national safety wage gap or causal effect.

Strict evidence 52 · Bounded 23 · Directional 677 · Counterexamples 7 · Conflicts 0

Recommended placement: Report Core Argument. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as supported. The reviewed corpus supports a bounded account of mechanisms and directional pressures affecting safety compensation more strongly than it supports any global wage-gap, prevalence, or causal-effect estimate. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

A national safety wage gap is not established

UNSUP-01 · Unsupported



Strict conclusion

A national safety wage-gap estimate is not supported by the current corpus.

Broader bounded conclusion

A national safety wage-gap estimate is not supported by the current corpus.

What I can responsibly say

A national safety wage-gap estimate is not supported by the current corpus.

What I cannot responsibly say

The corpus provides a national safety wage-gap estimate.

Strict evidence 0 · Bounded 0 · Directional 0 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Limitations. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as unsupported. A national safety wage-gap estimate is not supported by the current corpus. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

National mechanism prevalence is not established

UNSUP-02 · Unsupported

Strict conclusion

A national prevalence estimate for any compensation mechanism is not supported by the current corpus.

Broader bounded conclusion

A national prevalence estimate for any compensation mechanism is not supported by the current corpus.

What I can responsibly say

A national prevalence estimate for any compensation mechanism is not supported by the current corpus.

What I cannot responsibly say

The corpus estimates national mechanism prevalence.

Strict evidence 0 · Bounded 0 · Directional 0 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Limitations. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as unsupported. A national prevalence estimate for any compensation mechanism is not supported by the current corpus. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

A causal effect is not established

UNSUP-03 · Unsupported

Strict conclusion

A causal effect estimate for any compensation mechanism is not supported by the current design.

Broader bounded conclusion

A causal effect estimate for any compensation mechanism is not supported by the current design.

What I can responsibly say

A causal effect estimate for any compensation mechanism is not supported by the current design.

What I cannot responsibly say

The corpus identifies causal effects of compensation mechanisms.

Strict evidence 0 · Bounded 0 · Directional 0 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Limitations. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as unsupported. A causal effect estimate for any compensation mechanism is not supported by the current design. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

A regression result is not available

UNSUP-04 · Unsupported

Strict conclusion

No regression-based claim is supported because no regression-ready design passed the gate and no regression was run.

Broader bounded conclusion

No regression-based claim is supported because no regression-ready design passed the gate and no regression was run.

What I can responsibly say

No regression-based claim is supported because no regression-ready design passed the gate and no regression was run.

What I cannot responsibly say

Regression results establish the proposed relationship.

Strict evidence 0 · Bounded 0 · Directional 0 · Counterexamples 0 · Conflicts 0

Recommended placement: Appendix Only. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as unsupported. No regression-based claim is supported because no regression-ready design passed the gate and no regression was run. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

A fixed safety growth advantage is not established

UNSUP-05 · Unsupported

Strict conclusion

A precise claim that safety wages grow a fixed percentage faster is not supported by the available matched evidence.

Broader bounded conclusion

A precise claim that safety wages grow a fixed percentage faster is not supported by the available matched evidence.

What I can responsibly say

A precise claim that safety wages grow a fixed percentage faster is not supported by the available matched evidence.

What I cannot responsibly say

Safety wages grow X percent faster nationally.

Strict evidence 0 · Bounded 0 · Directional 0 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Limitations. Evidence tiers remain separate; conflicts do not support the claim.

I classify this claim as unsupported. A precise claim that safety wages grow a fixed percentage faster is not supported by the available matched evidence. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

Mechanisms do not prove a wage difference

UNSUP-06 · Unsupported

Strict conclusion

The evidence does not establish that a documented mechanism caused an observed wage difference.

Broader bounded conclusion

The evidence does not establish that a documented mechanism caused an observed wage difference.

What I can responsibly say

The evidence does not establish that a documented mechanism caused an observed wage difference.

What I cannot responsibly say

The documented mechanism caused the observed wage difference.

Strict evidence 0 · Bounded 0 · Directional 0 · Counterexamples 0 · Conflicts 0

Recommended placement: Report Limitations. Evidence tiers remain separate; conflicts do not support the claim.

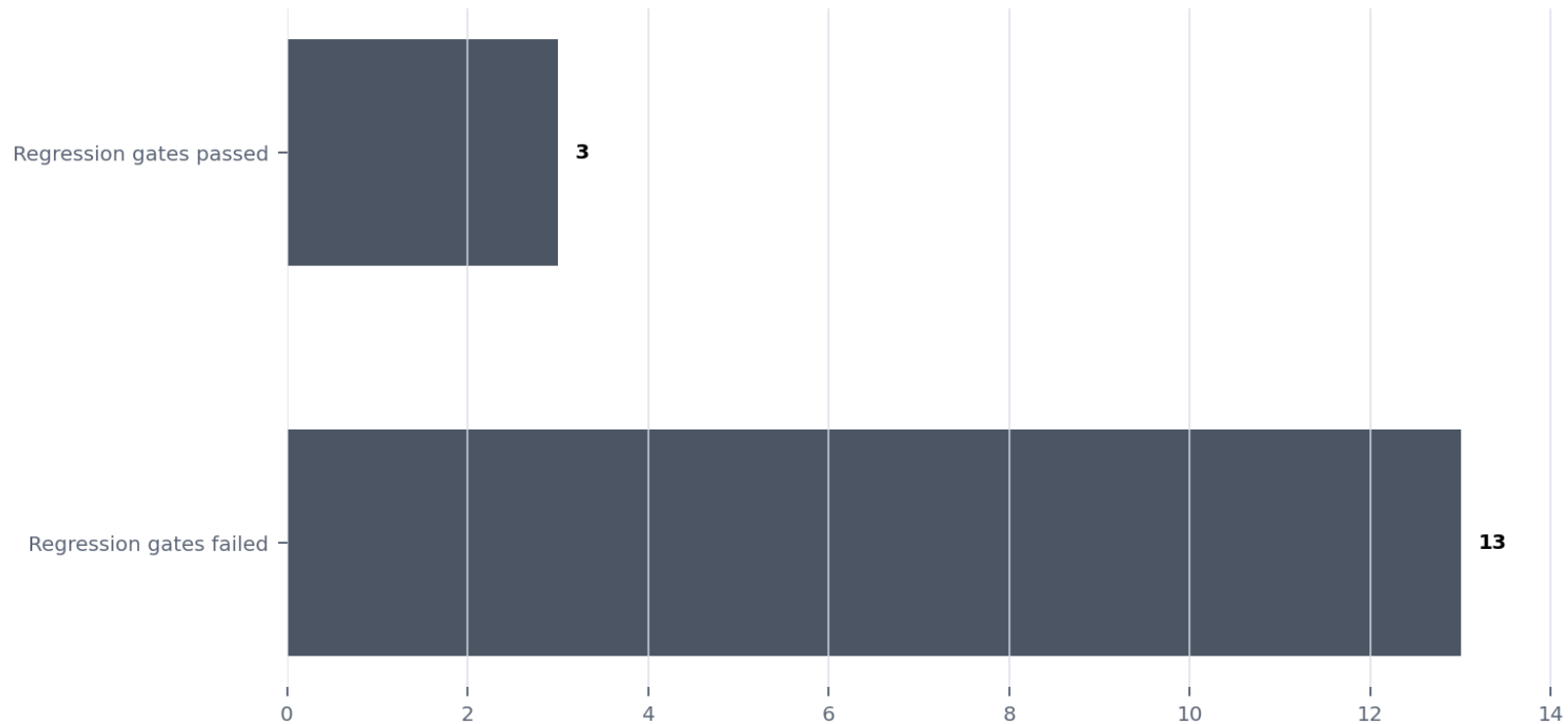
I classify this claim as unsupported. The evidence does not establish that a documented mechanism caused an observed wage difference. The broader evidence can add context or direction, but it cannot erase the strict evidence boundary. The main limit is unsearched, storage-held, conflict, linkage, compatibility, and independent-human-review limits remain. This card states what I can use in the report and what I must not turn into a national or causal conclusion.

Section III

What limited the analysis

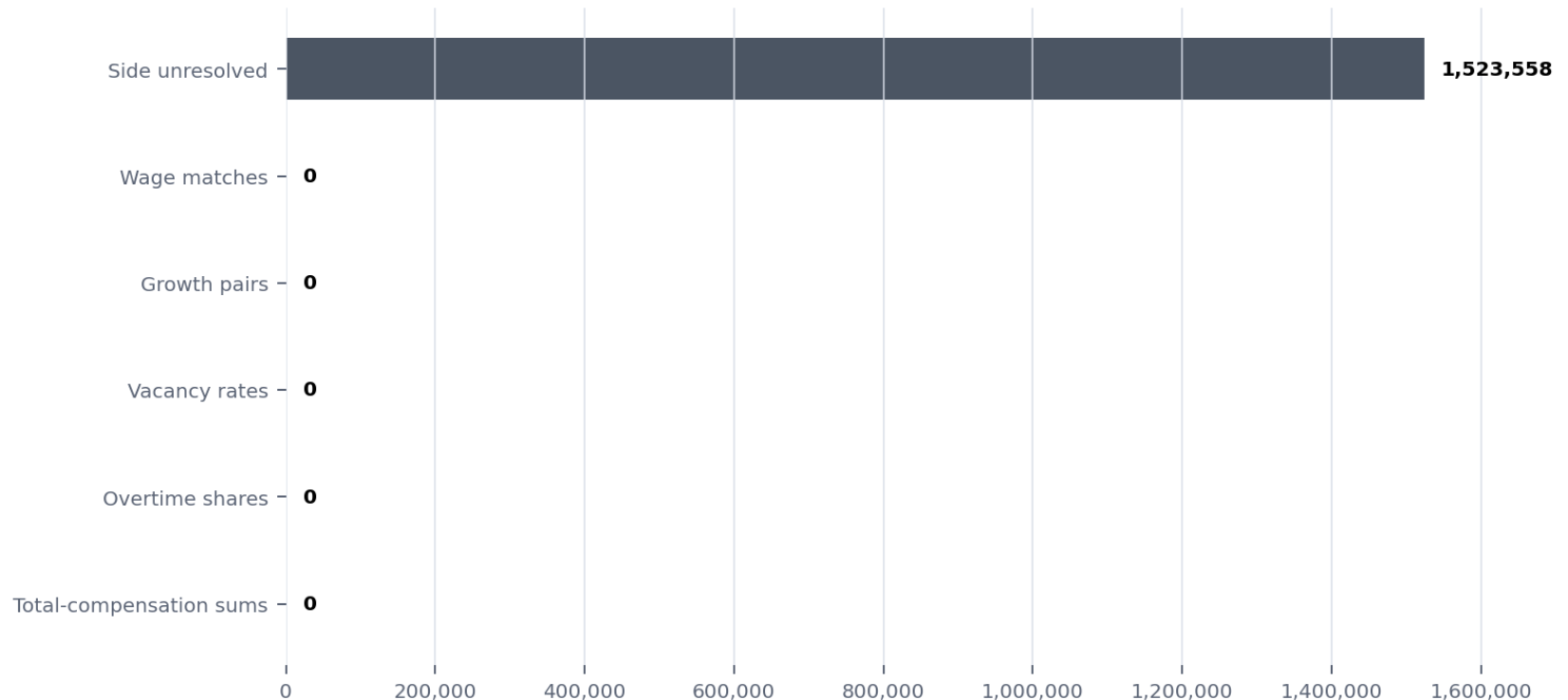
A visual section built from adjudicated, source-linked evidence.

Analytical-design limits



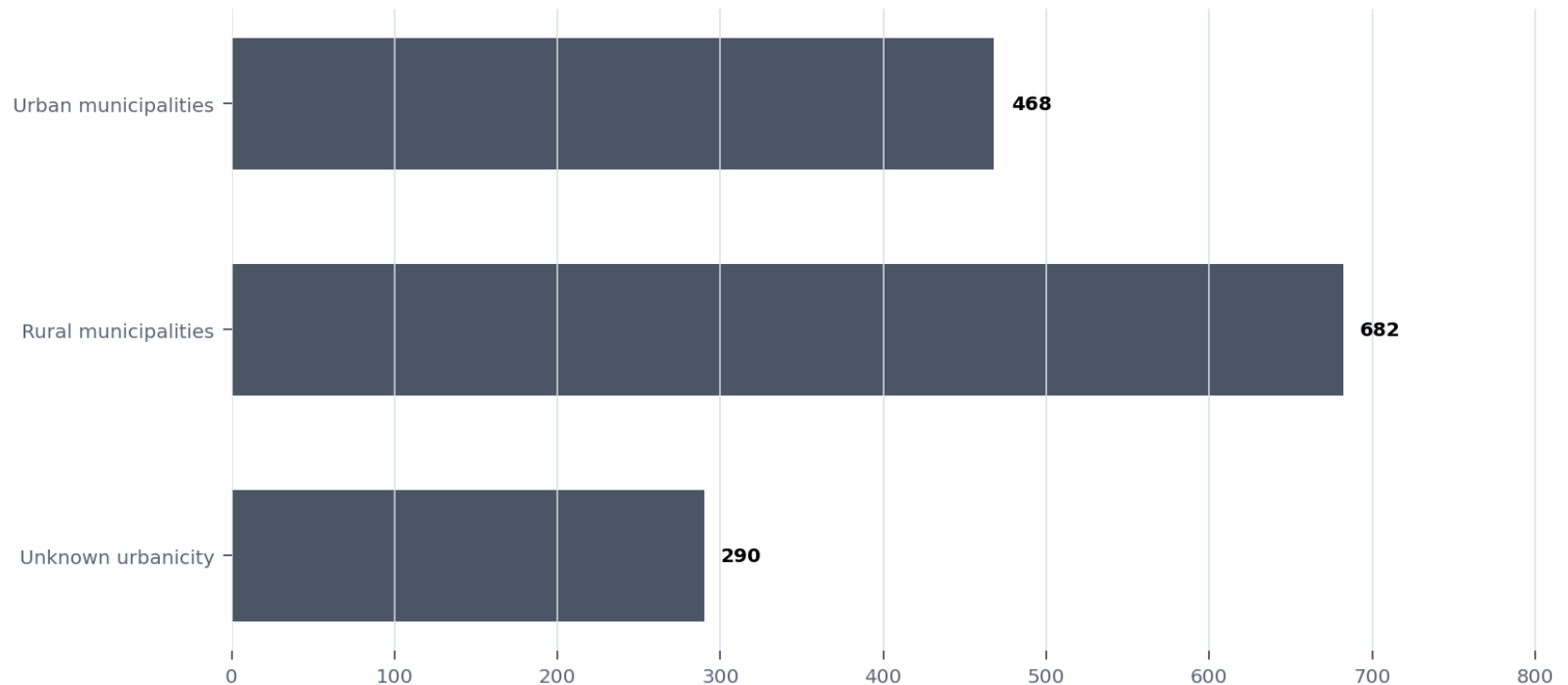
Only 3 of 16 regression-readiness gates passed. I therefore did not run a regression, estimate a national safety wage gap, estimate mechanism prevalence, or claim a causal effect. A larger row count would not repair unclear units, incompatible pay measures, missing controls, or repeated-event ambiguity. These are design failures, not merely small-sample problems. The correct result is therefore that the model was not design-ready, not a zero effect or evidence that the mechanisms do not matter.

Comparability limits



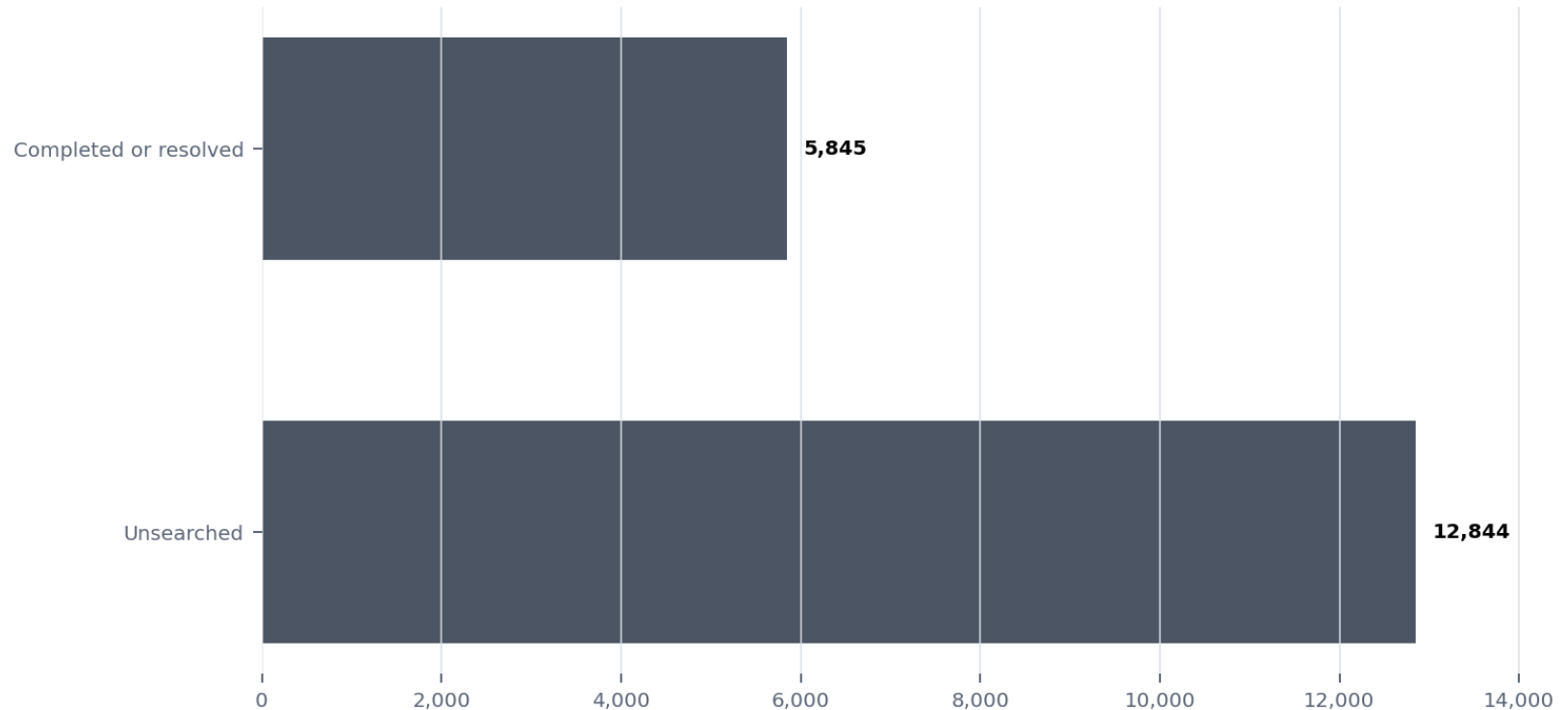
The project found extensive payroll, budget, and staffing material, but 1,523,558 reconciled observations still lacked a defensible safety or non-safety side. Pay basis, compensation basis, conflicts, identity, and timing created additional holds. That is why the external layer produced zero compatible wage matches, growth pairs, vacancy rates, overtime shares, or total-compensation sums. The constraint was compatible analytical units, not a lack of documents. Broader rules recovered local and directional evidence, but they still did not create a representative cross-occupation panel.

Coverage and representativeness limits



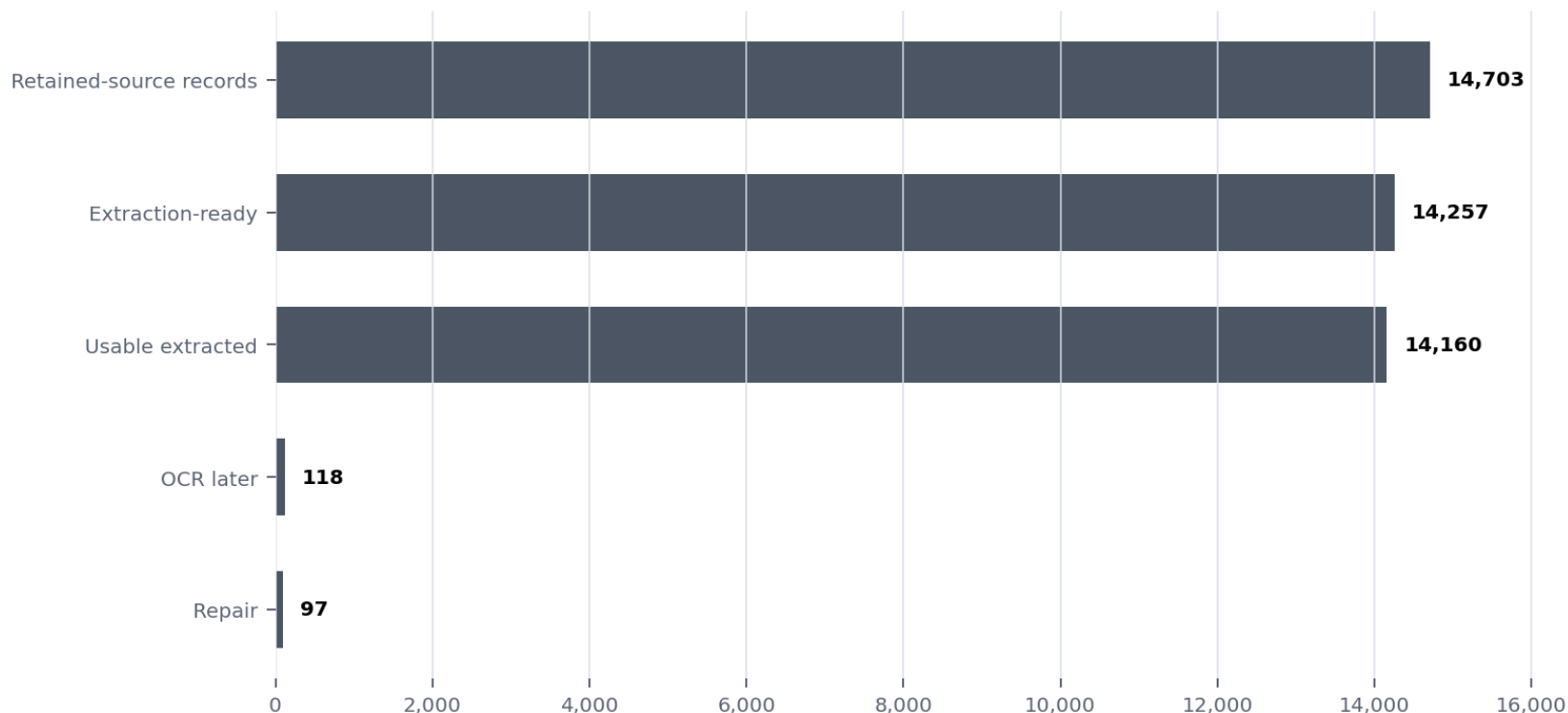
Geographic coverage is uneven across states and compensation mechanisms. The canonical crosswalk identifies 468 urban municipalities, 682 rural municipalities, and 290 with unknown urbanicity. Local comparisons remain local, and mechanism maps show where documented events appear—not how common a process is among all municipalities. Without an eligible-universe denominator, geographic density cannot support prevalence claims. The unknown and uneven cells stay visible rather than being inferred from neighboring municipalities or better-covered states.

Evidence discovery limits



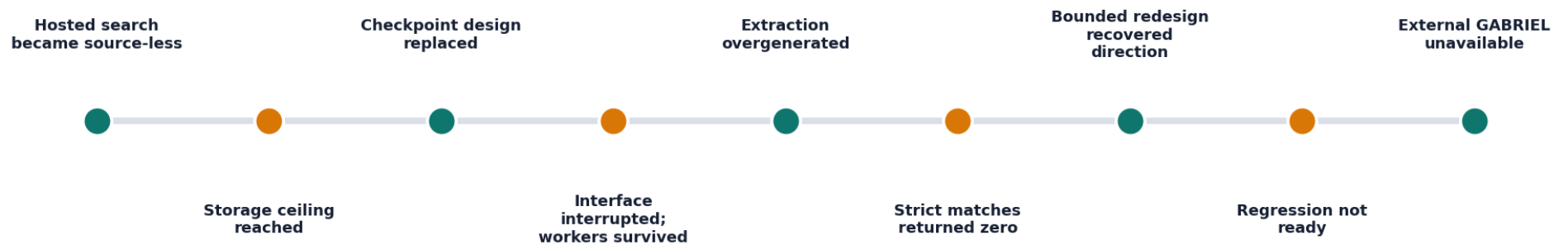
I began with 18,689 intended residual search targets. The local workflow completed or resolved 5,845, while 12,844 remained unsearched after the hosted-search backend stopped returning sources. Unsearched means unknown: it does not mean that no relevant evidence exists. This gap reduces completeness and prevents national generalization, but it does not erase findings supported by sources already retained and reviewed. The practical consequence is a ceiling on coverage: I can describe what the reviewed sources show, but I cannot assume the unseen targets would confirm the same pattern.

Extraction limits



The project inspected 14,703 retained-source records. Of 14,257 extraction-ready records, 14,160 produced usable extracted payloads; 118 PDFs were set aside for later OCR and 97 records required extraction repair. These are explicit availability limits, not silently imputed documents. The usable corpus is large, but claims that depend on the excluded records remain less complete. I kept these categories separate so a failed extraction could not be mistaken for a source that contained no relevant evidence.

Attempts that did not work—and what changed



The workflow was not linear. Hosted search became source-less, storage reached its ceiling, large checkpoints strained memory, and raw extraction overproduced noisy records. Strict matching then returned zero external wage and growth matches; a broader bounded redesign recovered usable directional evidence but did not upgrade any claim class. Regression still failed readiness, and external GABRIEL scoring did not run. Each failure changed the method while preserving earlier results. The repairs improved reliability and evidence use, but they could not recover information that the available sources did not contain.

What the evidence still supports

What the evidence supports

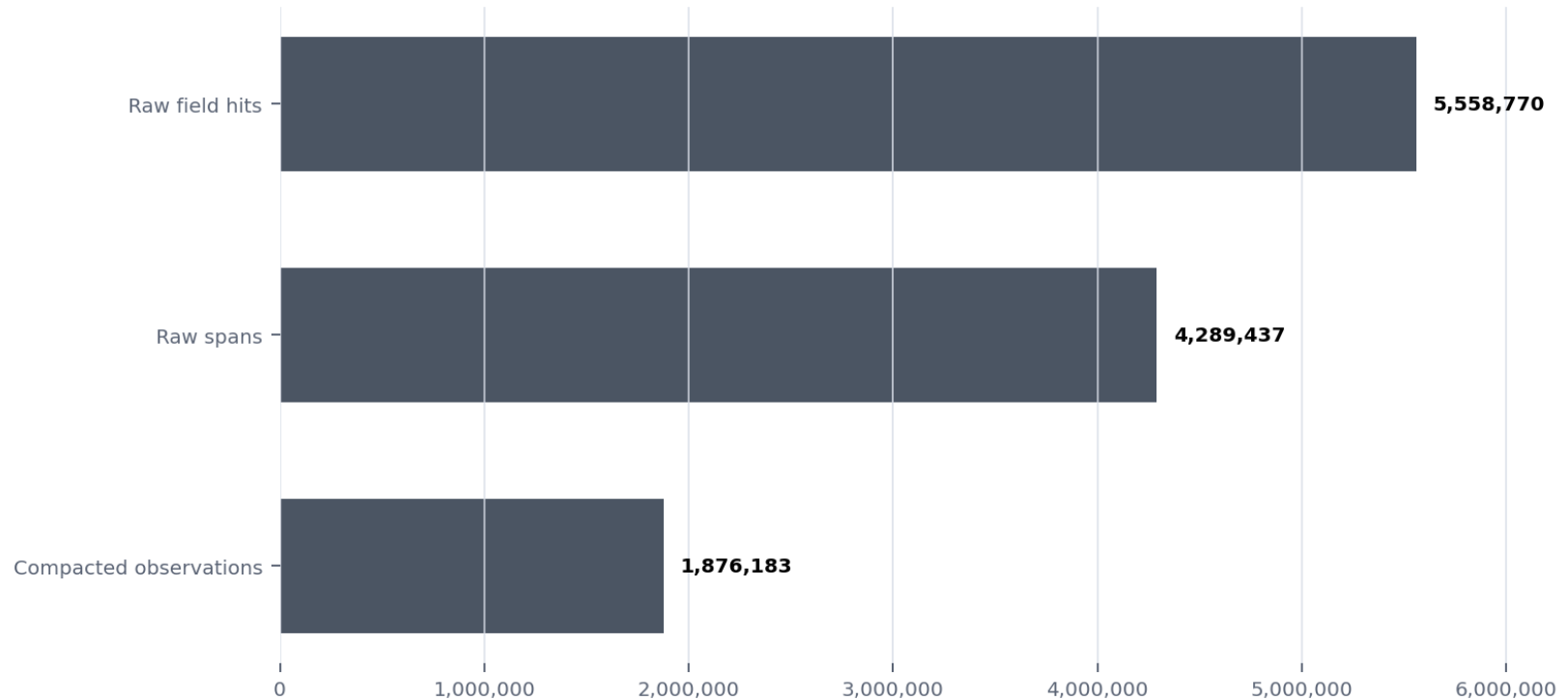
- Bounded mechanism claims
- Staffing and implementation patterns
- Local comparisons
- Documentary growth patterns
- Administrative corroboration

What it does not establish

- National wage gap
- Representative growth advantage
- Mechanism prevalence
- Regression estimate
- Causal effect

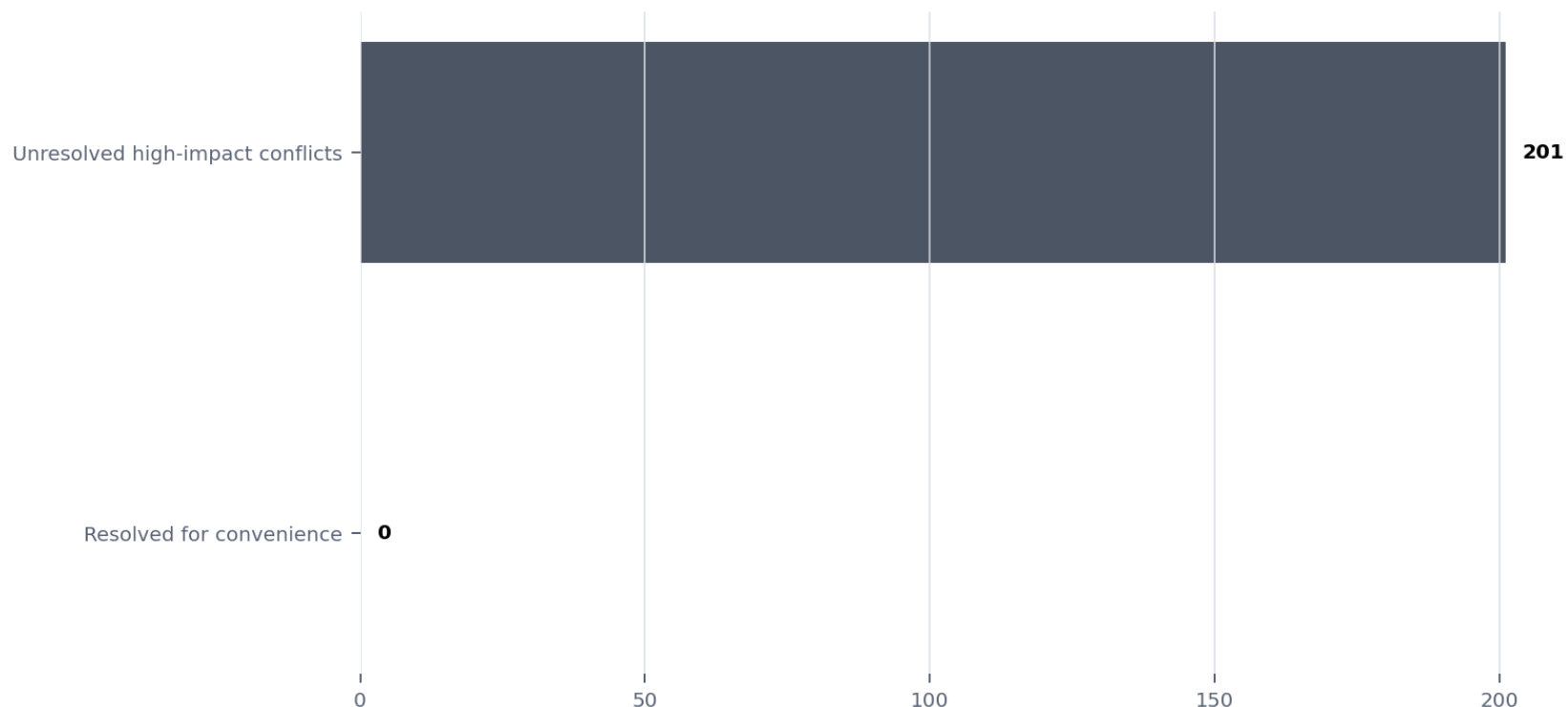
The final evidence supports one bounded claim, one conditional claim, five compensation-mechanism claims, a mixed finding, staffing and implementation patterns, local examples, documentary growth patterns, and administrative corroboration. It does not establish a national wage gap, a nationally representative safety growth advantage, mechanism prevalence, a regression estimate, or a causal effect. That boundary is the project's result, not an omitted final step. This distinction keeps the final package useful without letting a large corpus substitute for the comparison design the strongest claims require.

Raw extraction and compaction



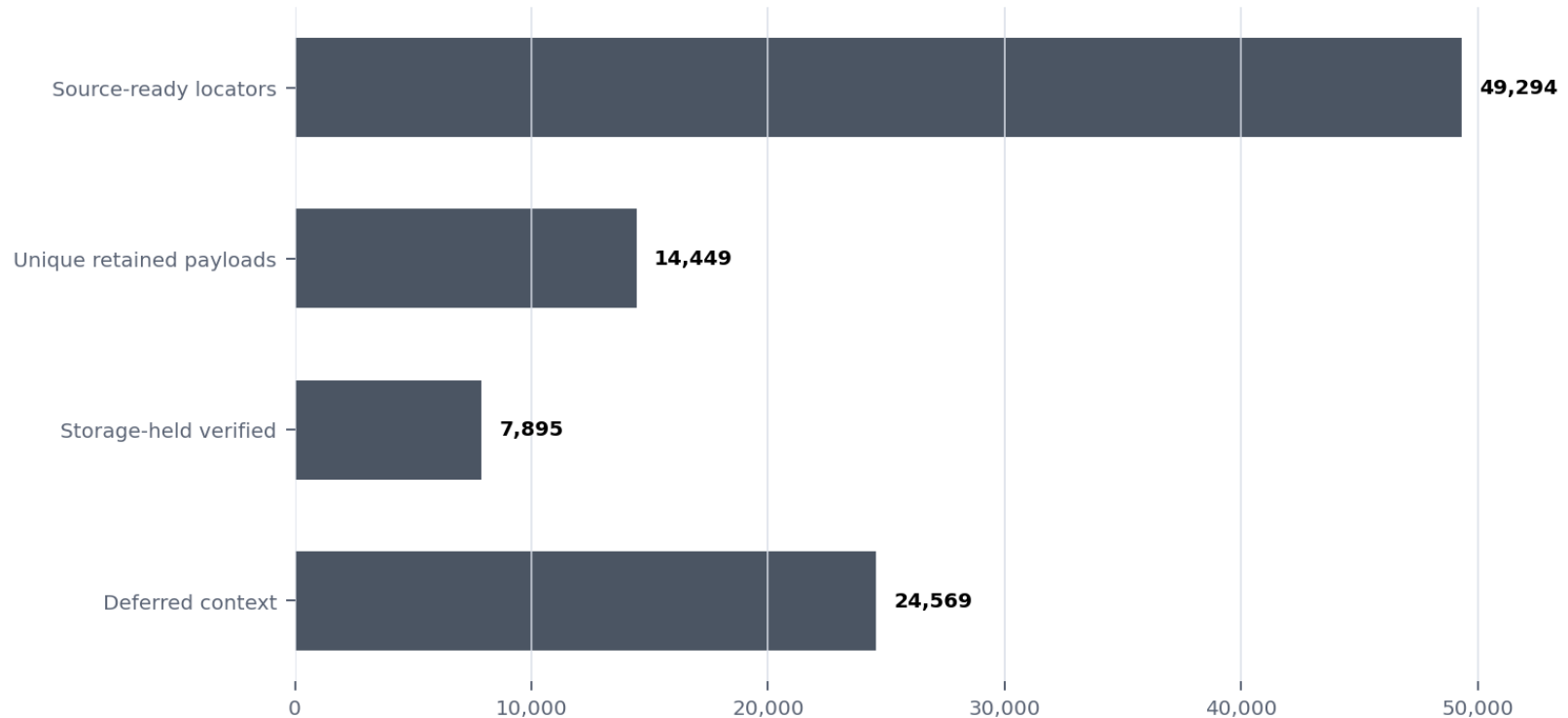
Automated field recovery produced 5,558,770 raw field hits and 4,289,437 raw spans. Compaction reduced that material to 1,876,183 administrative observations—a 66.25% reduction in field records—by removing duplication, boilerplate, and structurally unusable material. More extracted rows did not mean more independent facts. The compact observations still preserve many-to-many source lineage and cannot be read as event prevalence. This reduction was necessary to prevent repeated source fragments from inflating the apparent amount or strength of evidence.

Review limits



New administrative evidence was classified with deterministic local rules and bounded semantic AI review; it was not scored by GABRIEL. That review improves traceability but is not independent human gold coding. The project also preserved 201 unresolved high-impact conflicts rather than choosing convenient values. Those conflicts are visible as a limitation and excluded from clean headline calculations. Independent human review of the most consequential records would strengthen confidence, especially where wording or source version remains disputed.

Storage and source-retention limits



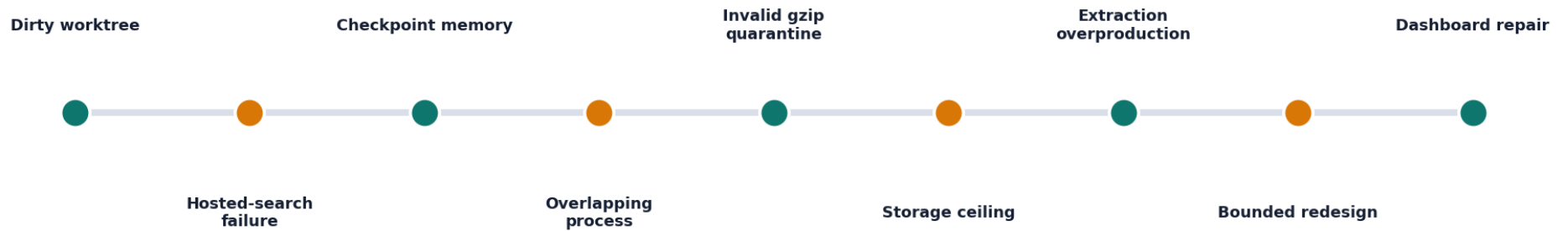
The review pipeline identified 49,294 source-ready locators, retained 14,449 unique payloads under a 30 GiB ceiling, and held 7,895 verified sources for possible later recovery. Another 24,569 low-value context sources were deferred. The storage limit therefore shaped which evidence could enter later stages. It does not indicate that held sources favor either side, and this atlas does not treat them as observed evidence. A targeted recovery remains possible later, but final adjudication found that no held-source tranche was required to produce the present bounded claims or visuals.

Section IV

How the project worked

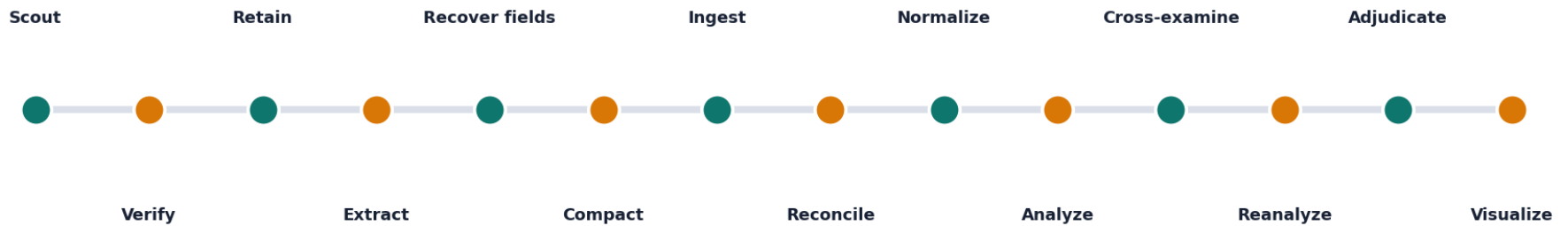
A visual section built from adjudicated, source-linked evidence.

Errors, interruptions, and recovery



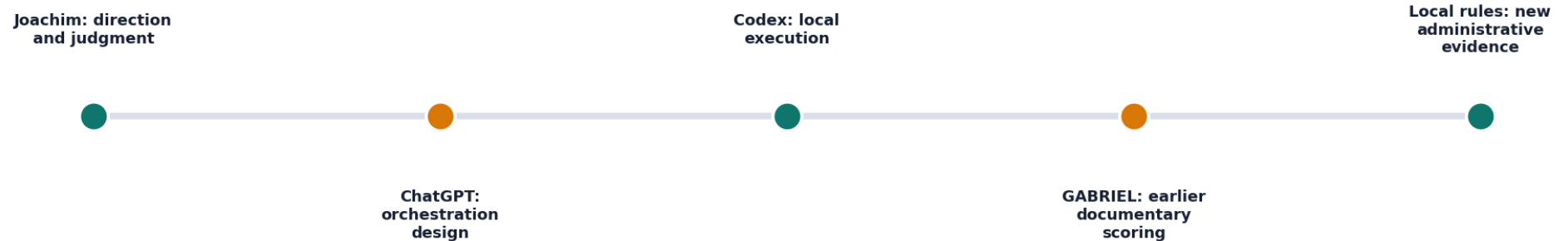
The pipeline encountered a dirty-worktree blocker, a source-less hosted-search backend, oversized checkpoints, overlapping preparation work, invalid gzip files, a storage ceiling, extraction overproduction, overly strict promotion gates, and dashboard deployment problems. The response was not to hide failed work. I preserved completed artifacts, quarantined invalid files, replaced fragile checkpoints, separated strict and bounded evidence, and repaired deployment. Some limits remained: no additional hosted search, no external GABRIEL scoring, and no defensible regression. The key recovery principle was to resume from accepted checkpoints and repair only the affected module. That preserved valid earlier work while preventing a failed source, process, or analytical rule from silently contaminating later outputs.

How the evidence moved through the project



I directed a staged workflow from scouting and verification through retention, extraction, compaction, reconciliation, mathematical analysis, semantic review, bounded reanalysis, claim adjudication, and visual production. Each stage narrowed the usable evidence while preserving source lineage and exclusions. The process is long because the research question requires matched occupations within the same city and compensation cycle. No later stage was allowed to turn unmatched administrative material into a clean wage comparison. At each transition, deterministic checks recorded the analytical unit, exclusions, denominator, source coordinates, and whether the evidence was strict, bounded, directional, contextual, or rejected. Those controls make the result reproducible while keeping later interpretation tied to the evidence that actually survived each gate.

Human and AI roles



I set the research question, evidence standards, priorities, corrections, and final judgment. ChatGPT helped design orchestration prompts, analytical frameworks, validation gates, and report structure. Codex executed local scripts, five-lane runs, checkpointing, validation, and artifact generation. GABRIEL scored earlier documentary evidence where available. When API capacity blocked new external scoring, deterministic local rules processed explicit administrative fields. The AI work remained bounded by my direction and is not a substitute for independent human validation. This division matters because the systems handled volume and reproducibility, while the research design, standards, and judgment remained human-directed. I did not manually review millions of records, and I did not accept model output without iterative correction and validation.

Strict and broader evidence lanes



I kept five evidence outcomes separate. Strict evidence supports exact, claim-safe statements. Bounded evidence permits compatible comparisons with explicit caveats. Directional evidence supports mechanisms or direction without a precise magnitude. Context records explain the setting, while rejected records do not support claims. The broader redesign recovered useful evidence without rewriting the strict baseline. It strengthened five claims within their existing class, made one claim more mixed, and produced no claim-class upgrades. The same admission rules applied to supporting and countervailing evidence. A broader record could strengthen context or mechanism interpretation, but directional evidence could not become a precise wage estimate and unresolved direction-changing conflicts remained excluded.

Technical notes and evidence boundary

Native PDF pages (1,029,482) remain separate from the 650,482 text-page equivalent for non-PDF material. The external administrative layer produced no compatible wage or growth matches. No regression, prevalence estimate, national wage-gap estimate, or causal estimate was produced. No hosted search, new network collection, OCR, held-source processing, or new claim adjudication occurred during atlas production.

Project-wide limits preserved

- 12,844 hosted-search targets remain unsearched.
- 7,895 verified sources remain storage-held.
- 201 high-impact conflicts remain unresolved.
- External administrative evidence was processed without new GABRIEL scoring.
- The semantic review was bounded and was not independent human gold coding.